

Chapter 22

Acute Pulmonary Embolism

Doubt is not a pleasant condition, but certainty is absurd.

Voltaire

Venous thromboembolism (i.e., deep vein thrombosis and pulmonary embolism) is cited as the third leading cause of cardiovascular deaths worldwide (behind myocardial infarction and stroke) (1), and has an annual risk of almost 10% in adults over the age of 45 (2). There is a heightened risk of VTE in ICU patients, which mandates the preventive measures for VTE described in [Chapter 5](#). This chapter focuses on the condition that is responsible for the life-threatening potential of VTE; i.e., acute pulmonary embolism.

There are two features of pulmonary embolism that deserve emphasis at the outset. First, pulmonary embolism is more common than suspected; e.g., autopsy studies in hospitalized patients report the presence pulmonary emboli in as many as 30% of patients (3), and in most of these cases, the condition was not suspected prior to death. Second, the clinical suspicion of pulmonary embolism is confirmed by diagnostic testing in as few as 10% of cases (4). These observations indicate that *pulmonary embolism suffers from both underdiagnosis and excessive diagnostic testing* (a rather unique pairing). This conundrum is adequately summarized in the following statement:

The one certainty surrounding the issue of thromboembolism diagnosis in critically ill patients is that considerable uncertainty remains (5).

The emphasis in this chapter is on acute pulmonary embolism that requires admission to an ICU, or develops during the ICU stay. Many of the recommendations in the chapter are based on clinical practice guidelines, which are listed in the bibliography at the end of the chapter (6–9).

CLINICAL EVALUATION

The diagnostic problems mentioned in the introduction are explained by the clinical manifestations of acute pulmonary embolism, which are neither sensitive nor specific (see next).

Clinical Presentation

The typical presentation of acute pulmonary embolism (acute PE) is a patient with dyspnea, tachycardia, and hypoxemia, but these are signs of cardiopulmonary disease, and do not implicate an acute PE. The common manifestations of acute PE are listed in [Table 22.1](#), along with the predictive value of each for the likelihood of disease ([10](#)). Note that none of the findings provides more than a 50% chance of identifying an acute PE (i.e., the positive predictive value), and that the absence of these findings does not exclude an acute PE (i.e., a negative predictive value of $\geq 98\%$ is used to exclude the presence of a condition). Of particular interest is the negative predictive value of 70% for hypoxemia, which means that *30% of patients with an acute PE have a normal arterial PO_2* . The alveolar–arterial PO_2 (A-a PO_2) gradient can also be normal in acute PE ([11](#)).

TABLE 22.1 Predictive Findings of Clinical Value in Acute Pulmonary Embolism

Findings	Positive Predictive Value [†]	Negative Predictive Value [‡]
Dyspnea	37%	75%
Tachycardia	47%	86%
Tachypnea	48%	75%
Pleuritic chest pain	39%	71%
Hemoptysis	32%	67%
Pulmonary Infiltrate	33%	71%
Pleural Effusion	40%	69%
Hypoxemia	34%	70%

[†]Positive predictive value is the percentage of patients with the finding who have a pulmonary embolus.

[‡]Negative predictive value is the percentage of patients without the finding who do not have a pulmonary embolus.

From Reference 10.

Less common manifestations of acute PE include fever (15% of cases), which is usually below 102° F ([12](#)), and syncope (10% of cases), which is often associated with right heart strain ([13](#)). However, neither of these manifestations is specific for acute PE. Also worthy of mention, acute PE can be the cause of an acute exacerbation of COPD in a small percentage (<10%) of cases ([14](#)).

Predisposing Factors

Because the clinical manifestations of acute PE are non-specific, the presence or absence of predisposing factors for venous thrombosis is used to determine the probability that an acute PE is present (and hence the need for diagnostic testing). These predisposing factors are presented in [Chapter 5](#), and will not be repeated here. However, all patients who spend more than a few days in the ICU should be considered at-risk for venous thrombosis (and hence acute PE), and this risk is especially high in postoperative patients and those with spinal cord injuries or multisystem trauma.

D-Dimer Assay

Active thrombosis is accompanied by some degree of clot lysis, and this produces cross-linked fibrin monomers, also called fibrin D-dimers, or simply *D-dimers*. Plasma D-dimer levels are elevated in the presence of acute thrombosis, and the D-dimer assay has become a popular tool for determining the probability of an acute PE.

Predictive Value

D-dimer levels have a positive predictive value of 27%, and a negative predictive value of 92% (15), which means that an elevated D-dimer level does not indicate a high probability of acute PE, but a normal D-dimer level is evidence against the presence of acute PE (and venous thrombosis). There are several conditions other than thrombosis that elevate D-dimer levels, including sepsis, inflammatory conditions, heart failure, renal failure, pregnancy, and advanced age (15). This would explain why a majority (up to 80%) of ICU patients have elevated plasma D-dimer levels (16), and for this reason, the D-dimer assay is not considered a useful diagnostic test in the ICU setting.

Summary

There are no clinical or laboratory findings that predict the presence of a pulmonary emboli with certainty, and the suspicion of acute PE is usually based on the presence of predisposing factors, and the absence of evidence for other conditions that could be responsible for the clinical presentation. Unfortunately, this is a flawed practice, since pulmonary emboli are found in as few as 10% of the diagnostic imaging studies obtained for suspected PE (4).

DIAGNOSTIC IMAGING

The imaging methods in Table 22.2 are used to confirm or exclude the diagnosis of acute PE. The following is a brief description of each method.

TABLE 22.2 Diagnostic Imaging Studies for Acute Pulmonary Embolism	
Study	Features
Venous Ultrasound	Positive: Reveals proximal leg DVT in 30–50% of cases of acute PE, which eliminates the need for other diagnostic tests. Negative: A negative result does not eliminate the possibility of acute PE.
CT Pulmonary Angiography (CTPA)	Positive: The method of choice for the detection of pulmonary emboli. Negative: The risk of contrast-induced nephropathy limits use when CrCL <30 mL/min/1.73 m ² . Radiation dose is also relatively high ¹ , which raises concerns about its use in pregnancy and breast feeding.
Ventilation-Perfusion Lung Scan	Positive: No radiocontrast dye, and has a lower radiation dose than CTPA ¹ . Negative: Diagnostic yield is limited in the presence of pulmonary infiltrates or chronic lung disease.

Echocardiography	Positive: Can be diagnostic for acute PE (i.e., mobile right-sided thrombi or McConnell's sign). Negative: Diagnostic findings are infrequent.
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¹See text for radiation dose.

Venous Ultrasound

Most pulmonary emboli originate from deep venous thrombosis (DVT) in the thigh and groin (6), so the evaluation of suspected PE can begin with a bedside ultrasound examination of the proximal leg veins. Venous ultrasound is advantageous in patients who are not suitable candidates for the other imaging methods (see later), and it is appealing in ICU patients because it eliminates the risk associated with transport to the radiology suite.

Methodology

Venous ultrasound is described in Chapter 1. There are two methods for identifying venous thrombosis. The principal method is *compression ultrasound*, where a compressive force is applied to a vein that is situated under the ultrasound probe. This normally compresses the vein and obliterates the lumen, (see Figure 1.7), but veins that are filled with thrombi will not compress. Therefore, an incompressible vein is used as evidence of venous thrombosis (17).

The *color Doppler mode* of ultrasound can be used as an adjunct to venous compression. This method converts flow velocities into color images, and the flow in arteries and veins can be identified by the direction of flow in relation to the ultrasound probe (see Figure 1.8). Venous thrombosis is then characterized by sluggish or absent flow in an incompressible vein. The combination of compression and color Doppler ultrasound is known as *duplex ultrasound*.

Test Performance

For the detection of proximal DVT in the legs, duplex ultrasound has a sensitivity $\geq 95\%$, a specificity $\geq 97\%$, a positive predictive value of 97%, and a negative predictive value of 98% (17). These numbers indicate that duplex ultrasound is a reliable method for the detection of proximal DVT in the legs.

Clinical studies have shown that 30–50% of patients with acute PE will have ultrasound evidence of proximal DVT in the legs (6,18). When DVT is detected by ultrasound, no further workup for pulmonary emboli is necessary (unless the patient is hemodynamically unstable), and anticoagulation is started using the same regimens used for acute PE. (Anticoagulation for DVT and PE differs only in the duration of treatment.) When venous ultrasound is unrevealing, another diagnostic imaging test is required.

Upper Extremity DVT

Thrombosis in the axillary and subclavian veins can develop as a result of indwelling central venous catheters. Upper extremity DVT does not frequently result in symptomatic acute PE (<10% of cases) (19), but inspection for swelling of the upper arms is warranted in patients with indwelling central venous catheters. Venous ultrasound is reliable for the detection of upper extremity DVT, with a sensitivity of 97% and a specificity of 96% (19). A positive study warrants anticoagulation and immediate removal of the responsible catheter (Table 22.2.).

CT Pulmonary Angiography

The diagnostic test of choice for acute PE is computed tomographic pulmonary angiography (CTPA) (6–9), which has replaced conventional pulmonary angiography. This is a specialized method of CT that uses a spiral or helical scanner that rotates around the patient to produce a “volumetric” two-dimensional view of the lungs. When multidetector scanners are used, peripheral injection of a radiocontrast agent allows visualization of the pulmonary arteries down to the subsegmental level. Pulmonary emboli appear as filling defects, as shown in Figure 22.1.

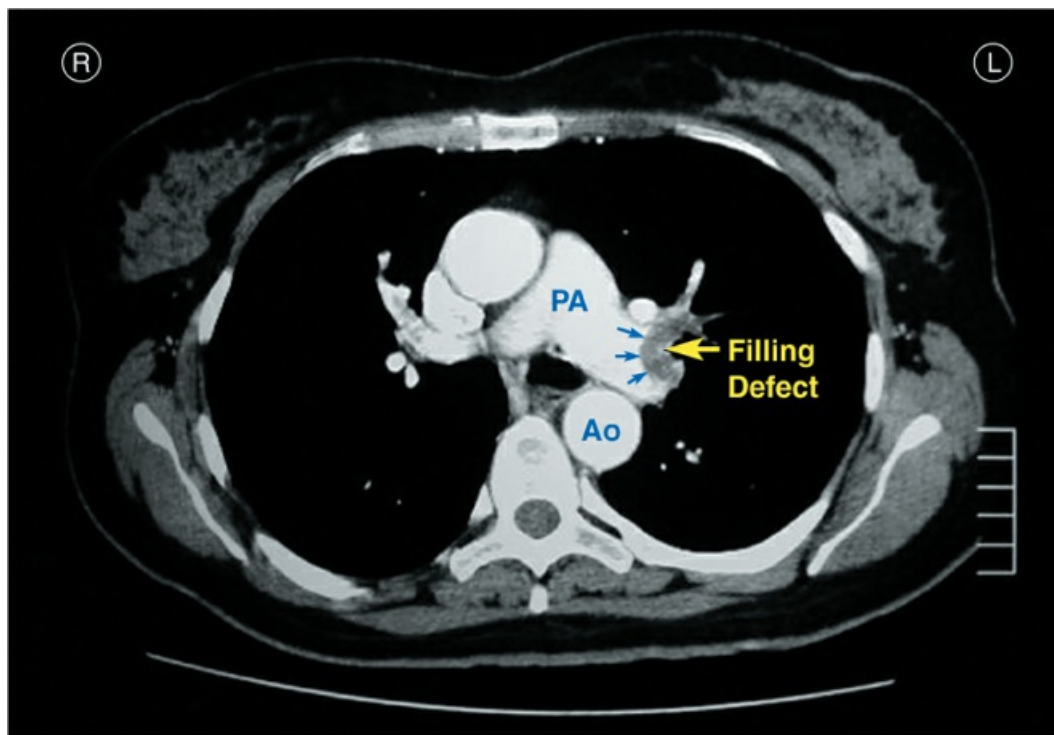


FIGURE 22.1 CT pulmonary angiogram showing a pulmonary embolus, which appears as a filling defect in the left main pulmonary artery. AO = aorta, PA = pulmonary artery. Image digitally retouched

Multidetector CTPA has a negative predictive value of 95% (20), which means a normal CTPA essentially eliminates the possibility of acute PE.

Subsegmental Emboli

The detection of pulmonary emboli on CTPA usually prompts immediate anticoagulant therapy, but there is some evidence that small emboli in subsegmental arteries (<2.5 mm in diameter) may not require anticoagulation. An analysis of the pooled results of 14 clinical studies showed that withholding anticoagulation in patients with subsegmental emboli and no evidence of DVT had no significant influence on mortality rate or recurrence rate for PE (21). However, a more recent study showed that withholding anticoagulation was not detrimental if there was a single subsegmental embolus, but if there were multiple subsegmental emboli, the recurrence rate for PE was much higher without anticoagulation (22).

WHAT TO DO: Based on the available studies, it seems reasonable to withhold anticoagulation for

subsegmental pulmonary emboli if the following conditions are satisfied: 1) there is a single subsegmental embolus, 2) there is no residual clot burden in the legs (by venous ultrasound), 3) there is no evidence of RV dysfunction, and 4) there are no significant co-morbidities (e.g., hypoxemia).

Adverse Effects

The major concern with CTPA is the risk of nephrotoxicity from the radiocontrast dye. Predisposing factors include renal insufficiency, diabetes, and volume depletion. The current recommendation is to avoid radiocontrast dye, if possible, when renal function is unstable, or the creatinine clearance is $<30 \text{ mL/min/1.73 m}^2$ (23). If CTPA is considered necessary, protection from dye-induced nephropathy is possible with a regimen of volume infusion and intravenous N-acetylcysteine (24). (See Chapter 34 for more on dye-induced nephropathy.) CTPA is also contraindicated in patients with a history of dye-induced anaphylaxis.

RADIATION EXPOSURE: An undesirable feature of CTPA that is often overlooked is the relatively high radiation exposure, which averages 15 millisieverts (mSv) per procedure (25). To place this in perspective, the average radiation exposure from a chest x-ray (single view) is 0.02 mSv (25), and the annual radiation limit for workers is 50 mSv (26). Because of the radiation dose, alternatives to CTPA should be considered during pregnancy and breast feeding (6). (Appendix 2 contains a table that lists the radiation dose of common radiographic procedures.)

Lung Scintigraphy

Radionuclide ventilation-perfusion scans (V/Q scans) were popular prior to the introduction of CT pulmonary angiography. However, they are no longer favored because the presence of lung disease (particularly infiltrative disease) produces abnormal scans. As a result, V/Q scans have diagnostic value in only 20–30% of cases (27). They are most reliable in younger patients with no underlying lung disease (which excludes most ICU patients, and have a much lower radiation exposure ($\sim 2 \text{ mSv}$) than CTPA studies.

The major role of V/Q scans is in cases of suspected PE where CT angiography is not performed (e.g., because of renal failure or a history of dye-induced anaphylaxis). V/Q scans have three possible results (6):

- . A normal V/Q scan has no ventilation or perfusion defects, and excludes the diagnosis of acute PE.
- . A high probability V/Q scan has perfusion defects without associated ventilation defects, and confirms the diagnosis of acute PE.
- . An indeterminate V/Q scan has matching ventilation and perfusion defects, and can neither exclude nor confirm the diagnosis of acute PE.

About three-quarters of V/Q scans are indeterminate, which has no diagnostic value, but the combination of an indeterminate scan and a negative venous ultrasound study can be used to exclude the diagnosis of acute PE, but only in patients who are hemodynamically stable (6).

Echocardiography

Transthoracic echocardiography is typically obtained after the diagnosis of acute PE has been confirmed, to look for signs of right ventricular (RV) dysfunction (see [Figure 18.2](#)). Evidence of right heart strain (i.e., RV dilation with elevated troponin levels) has prognostic significance in acute PE, but has no diagnostic significance (i.e., is not specific for PE).

Diagnostic Aids

There are two findings on transthoracic cardiac ultrasound that are specific for acute PE:

- . Mobile thrombi in the right heart have been reported in 2–18% of patients with acute PE ([28,29](#)), and are more prevalent in cases of massive PE ([28](#)).
- . Akinesis of the free wall of the RV, combined with normal motion at the apex (called McConnell’s sign) is considered specific for acute PE ([30](#)), and is reported in 20% of cases ([28](#)).

These two findings are relatively infrequent, but they add diagnostic value to cardiac ultrasound in cases where the other imaging modalities are nondiagnostic.

MANAGEMENT

The early management of acute PE is organized according to a risk stratification system (based on the risk of an unfavorable outcome) that has three levels of illness severity:

- . Low-risk PE: No right ventricular dysfunction, hemodynamic instability, or significant co-morbid conditions (e.g., hypoxemia).
- . Intermediate-risk PE: (Also known as submassive PE) Evidence of right ventricular dysfunction (with or without elevated troponin levels), but no hemodynamic instability.
- . High-risk PE: (Also known as massive PE.) Evidence of right heart strain (RV dysfunction plus elevated troponin levels) with hemodynamic instability.

The antithrombotic strategies for each risk level are summarized in [Table 22.3](#).

Risk Category	Recommendations
Low Risk	1. Start oral anticoagulation with either: a. Apixaban: 10 mg BID x 7 days, then 5 mg BID. b. Rivaroxaban: 15 mg BID x 3 weeks, then 20 mg once daily. 2. If CrCL ≤ 30 mL/min/1.73 m², use warfarin, and anticoagulate with reduced-dose enoxaparin (1 mg/kg once daily) until the INR is therapeutic.
Intermediate Risk	1. Start anticoagulation with enoxaparin (1 mg/kg subQ every 12 hrs). 2. If no deterioration after 24–48 hrs, switch to an oral anticoagulant using the recommendations for low-risk PE.
High Risk	1. Start anticoagulation with unfractionated heparin, using the weight-based regimen below: a. Give IV load of 80 IU/kg. then infuse at 18 U/kg/hr.

- b. Check PTT at 6 hrs, and adjust dose to a target PTT of 46–70 sec.
2. Initiate thrombolytic therapy as soon as possible (if no contraindications). The recommended drug regimens are:
 - a. Alteplase (tPA): 100 mg infused over 2 hrs.
 - b. For impending cardiovascular collapse, give 0.6 mg/kg over 15 minutes (max = 50 mg).
3. If the condition does not improve, consider catheter-directed thrombolysis or embolectomy.

Low-Risk PE

Oral anticoagulation (started as soon as possible after the diagnosis) is suitable for low-risk PE, and the direct-acting oral anticoagulants (DOACs) are preferred to warfarin for reasons explained in [Chapter 19](#). The drug regimens with documented success in treating venous thromboembolism are *rivaroxaban* in a dose of 15 mg twice daily for 3 weeks, then 20 mg once daily ([31](#)) and *apixaban* in a dose of 10 mg twice daily for 7 days, then 5 mg twice daily ([32](#)). The starting dose for both of these drugs is higher than usual (see [Table 19.3](#)), but there is no increase in the risk of bleeding.

DOACs are not recommended when renal function is impaired and the creatinine clearance is ≤ 30 mL/min/1.73 m² ([6](#)). (Note: 1.73 m² is the body surface area of an average-sized adult.) In this situation, warfarin is the recommended oral anticoagulant. However, at least 2–3 days are required to achieve therapeutic anticoagulation after starting warfarin, and reduced-dose enoxaparin (i.e., 1 mg/kg once daily) is recommended for maintaining anticoagulation until the INR reaches the therapeutic range ([6](#)). Enoxaparin can be used down to a CrCL of 15 mL/min ([33](#)).

Finally, since oral anticoagulants are used for low-risk PE, and the drug effect is not monitored, patients can be discharged (from the emergency department or the hospital) if there are no significant co-morbidities (e.g., hypoxemia).

Intermediate-Risk PE

Because patients with intermediate-risk PE have some degree of RV dysfunction, and may have right heart strain (i.e., RV dysfunction plus elevated troponins) ([34](#)), they are monitored in the hospital for 24–48 hours for signs of hemodynamic deterioration. During this time, enoxaparin is recommended for anticoagulation ([6](#)), using a dose of 1 mg/kg every 12 hours. If there is no deterioration after the observation period, oral anticoagulation can be started, using the drugs and dosing regimens recommended for low-risk PE.

Thrombolytic Therapy

Thrombolytic therapy is not officially recommended for intermediate-risk PE, but it has been used (by this author and others) in cases of severe RV dysfunction or unresolving right-heart strain (i.e., RV dysfunction plus elevated troponin levels). Studies in patients with PE and right heart strain have shown that thrombolytic therapy reduces the risk of cardiovascular decompensation, but it also increases the risk of serious bleeding, including intracranial hemorrhage (see later). The risk of bleeding, more than lack of a hemodynamic benefit, has discouraged the use of thrombolytic therapy in hemodynamically stable cases of PE.

High-Risk PE

High-risk PE is a life-threatening condition, with severe RV dysfunction leading to

hemodynamic instability. The principal goals of management are reperfusion of the pulmonary circulation and hemodynamic improvement.

Anticoagulation

Unfractionated heparin (UFH) is recommended for early anticoagulation in high-risk patients using the weight-based dosing regimen in [Table 22.3](#) (which uses actual body weight) (6). Weight-based dosing achieves more rapid anticoagulation than fixed dosing (35). As always with unfractionated heparin, therapy is aimed at achieving an activated partial thromboplastin time (PTT) of 46–70 seconds, or 1.5 to 2.5 times control. (Most ICUs have a heparin dosing guideline.) Heparin is continued through thrombolytic therapy until the condition improves or stabilizes.

HEPARIN DOSING IN OBESITY: The popular weight-based dosing regimen for heparin was derived in patients weighing less than 130 kg (286 lbs). For body weights in excess of 130 kg, this regimen will overestimate the heparin dose and promote excessive anticoagulation (36). To avoid this problem, the adjusted body weight shown here is recommended for weight-based heparin dosing in patients with morbid obesity (i.e., a body mass index ≥ 40 kg/m²) (37).

$$\text{Adjusted Weight (kg)} = \text{IBW} + 0.4 \times (\text{actual} - \text{ideal body weight}) \quad (22.1)$$

where IBW = ideal body weight. (Appendix 2 has a table of ideal body weights in adults). The adjusted body weight is almost halfway between the ideal and actual body weights.

Thrombolytic Therapy

Thrombolytic therapy is recommended for all patients with high-risk PE and hemodynamic instability. Absolute contraindications to thrombolytic therapy include active bleeding, a prior hemorrhagic stroke, an ischemic stroke within the past 6 months, central nervous system neoplasm, and major trauma, surgery, or head injury in the past 3 weeks (6).

DRUG REGIMEN: The recommended drug regimen is alteplase (tPA), 100 mg infused over 2 hours, and an accelerated regimen (0.6 mg/kg over 15 minutes to a maximum dose of 50 mg) is recommended if there is impending (or actual) cardiovascular collapse (6).

Heparin is used in conjunction with thrombolytic therapy, and the infusion can be continued during the thrombolytic treatment period (although it is frequently stopped and restarted after the fibrinolytic agent is given). Heparin therapy is particularly advantageous after thrombolysis because clot dissolution releases thrombin, and this can promote thrombotic reocclusion of the involved vessel.

SYSTEMIC VS. DIRECTED THROMBOLYSIS: Traditional (systemic) thrombolytic therapy provides some hemodynamic benefit (e.g., a decrease in pulmonary artery pressures), but there is a 10–12% incidence of major bleeding (including intracranial hemorrhage in 1–2% of cases) (38,39), and this erases the overall benefit of the treatment. Catheter-directed thrombolysis (through a catheter placed in the pulmonary artery) uses a lower thrombolytic dose, and causes fewer major bleeding episodes than systemic thrombolysis (40). (The effects on pulmonary artery pressures are similar to systemic thrombolytic therapy.) The major shortcoming of catheter-directed

thrombolysis is availability (an interventional radiologist is needed to place the catheter under fluoroscopic guidance), especially since the benefit of thrombolytic therapy is time-dependent; i.e., the greatest benefit occurs within 48 hours of symptom onset (6).

Mechanical Thrombectomy

Clot removal from the pulmonary arteries is possible using specialized catheters that are inserted percutaneously into the femoral vein and advanced into the pulmonary arteries. The emboli are located by contrast injection, and are then removed by aspiration. The clot burden that can be removed with this procedure is shown in [Figure 22.2](#).

Mechanical thrombectomy can result in dramatic improvements in pulmonary artery pressures and RV function (much greater than seen with thrombolytic therapy), and these changes persist for at least three months after the procedure (41). Although it is currently recommended as a rescue procedure when thrombolytic therapy fails or is contraindicated (6), it could be used as a replacement for thrombolytic therapy, if available. In fact, a look at the amount of clot removed from the pulmonary arteries in [Figure 22.2](#) provides an explanation of why thrombolytic therapy has had such a limited impact in life-threatening pulmonary embolism.

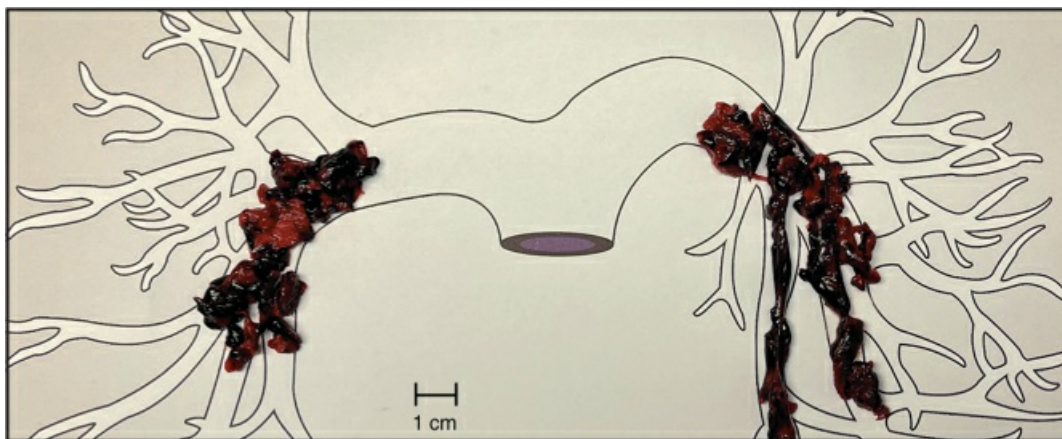


FIGURE 22.2 Catheter-based clot retrieval from the pulmonary arteries in a patient with acute pulmonary embolism and severe right heart strain. Image courtesy of Dr. Brian Shaw, DO (digitally retouched).

Hemodynamic Management

The management of cardiovascular compromise from an acute PE is challenging, because the primary problem is a physical obstruction in the pulmonary arteries. The following are some relevant points about management:

- . Aggressive volume infusion can be detrimental in acute PE because it can increase volume overload in the right ventricle, which will push the interventricular septum into the left ventricle and impede left ventricular filling (see [Figure 18.2](#)). Because of this risk, cardiac ultrasound should be used to guide fluid management in acute PE.
- . In the setting of right heart failure, a normal central venous pressure (4–6 mm Hg) can be used as evidence of hypovolemia (and the subsequent need for volume infusion). However, the volume infused should be limited to 500 mL (6), because of the previously mentioned risk. Once again, cardiac ultrasound should be used to guide volume infusion.

- . Current guidelines recommend dobutamine as a consideration when there is right heart strain and the blood pressure is normal (i.e., intermediate-risk PE) (6). Dobutamine is a vasodilator, and should not be used alone if acute PE results in hypotension. (See [Table 16.3](#) for the dobutamine dosing regimen.)
- . Norepinephrine is recommended for hypotension in acute PE because it does not increase pulmonary vascular resistance (6).

Vena Cava Filters

The aim of anticoagulation in acute PE is not to dissolve the existing embolus, but to prevent the next one. When anticoagulation is not possible, or is not effective, meshlike filters can be placed in the inferior vena cava (IVC) to prevent recurrent PEs from leg veins (42).

Filter Design

Inferior vena cava (IVC) filters have progressed through several iterations. The earliest filters were shaped like an umbrella, and were problematic because as they filled with trapped blood clots, they would obstruct the IVC and create troublesome leg edema. The design that led to the popularity of IVC filters is shown in [Figure 22.3](#). The design benefit is an elongated, conical shape (like a badminton birdie), which allows the basket to fill to 75% of its capacity without compromising the cross-sectional area of the vena cava. This limits the risk of vena cava obstruction and leg edema.

IVC filters are inserted percutaneously into the internal jugular vein or femoral vein, and are placed below the renal veins, if possible. The early filters (like the one in [Figure 22.3](#)) were permanent, but retrievable filters were introduced in 2003 (43), and the most recent iteration is a biodegradable filter with a basket that dissolves after 60 days (44).

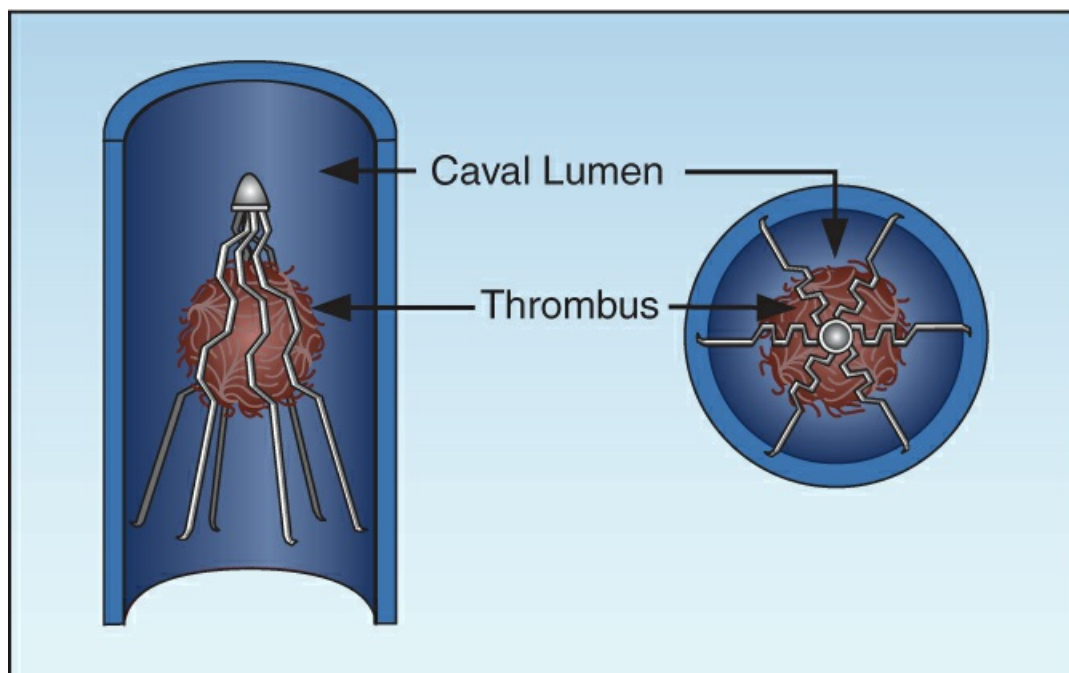


FIGURE 22.3 The Greenfield™ vena cava filter (Boston Scientific, Malborough, MA). The elongated, conical shape allows the basket to trap blood clots without obstructing blood flow,

and the struts have hooked ends to anchor the filter to the wall of the vena cava.

Indications

The consensus indications for placement of an IVC filter are as follows:

- . The presence of a DVT or acute PE and an absolute contraindication to anticoagulation (e.g., from active bleeding).
- . An acute PE that occurs while receiving therapeutic doses of an anticoagulant, and the likely source is DVT in the legs or pelvis (and not in an upper extremity).

IVC filters are also used as a preventive measure in conditions with a high risk of venous thromboembolism, such as multisystem trauma or bariatric surgery. Clinical studies have shown that the prophylactic use of IVC filters reduces the risk of acute PE by about 60%, but there is also a 70% increase in the incidence of DVT (45), so the overall benefit of the practice is questioned (6).

The Clinical Experience

Although not devoid of risk, IVC filters are remarkably safe and effective. The incidence of symptomatic PE after filter placement is about 5% (46), and life-threatening complications (e.g., migration of the filter) are reported in less than 1% of cases (46). One of the intriguing features of IVC filters (and one that is rarely mentioned) is that they never seem to get infected, even when exposed to bacteremia.

A FINAL WORD

An Exercise in Diagnostic Uncertainty

The experience with venous thromboembolism, which is far from satisfying, can be summarized as follows:

- . When you think it's there, it usually isn't; i.e., when a pulmonary embolism is suspected, the diagnosis is confirmed in as few as 10% of cases.
- . When it is there, you usually don't know it; i.e., in most cases of pulmonary embolism, the source of the embolus is clinically silent.
- . The best thing to do, then, is to prevent it (as described in Chapter 5).

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Chapter 23

Asthma and COPD in the ICU

Asthmatic distress calls for procedures which stimulate the sympathetic and dilate the bronchi.

Lawrason Brown, MD ([a](#))

Admission to an ICU is an ominous sign for patients with acute exacerbations of asthma and chronic obstructive pulmonary disease (COPD). For COPD patients, one of every four will not survive the hospitalization, and 40% of the survivors will be readmitted within a year ([1](#)). And for patients with asthma, the need for mechanical ventilation has been identified as the strongest risk factor for death ([2](#)).

This chapter describes the management of acute exacerbations of asthma and COPD, including some relevant issues related to positive pressure ventilation in patients with severe airflow obstructions. As customary, the recommendations in this chapter will follow the most recent clinical practice guidelines, which are listed in the bibliography at the end of the chapter ([3–6](#)).

One point of interest deserves mention before proceeding. The management of asthma and COPD is centered on dilating the airways with adrenergic agents, which is very similar to the statement in the introductory quote. The interesting point is that this statement was made almost 100 years ago, in 1931 ([a](#)). The significance of this is left to the judgment of the reader.

BASICS

Monitoring Airway Obstruction

The management of obstructive airways disease is guided by the severity of obstruction in the airways.

Clinical Evaluation

During acute exacerbations of asthma or COPD, *the clinical examination is often unreliable for determining the severity of airway obstruction* ([7](#)). There are usually signs of respiratory distress, including tachycardia, tachypnea, use of accessory muscles of respiration, and inability to

complete sentences without taking a breath, and there may be altered mentation and pulsus paradoxus (i.e., an inspiratory drop in systolic pressure >12 mm Hg). However, none of these clinical findings is sensitive or specific for severe airways obstruction, and more objective measures are recommended (3).

Measures of Peak Expiratory Flow

For spontaneously breathing patients, a readily obtainable bedside measure of airway obstruction is the peak expiratory flow rate (PEFR) (3). The PEFR varies with age, gender, and height, and measurements are typically expressed as “% predicted” (observed result/predicted result $\times$ 100), with the predicted values derived from standard reference equations. (Appendix 2 contains tables of the predicted PEFR for age, gender, and height.) The correlations between the % predicted values and the severity of airway obstruction are shown below (3).

<u>% Predicted PEFR</u>	<u>Severity of Obstruction</u>
$\geq 70\%$	Mild
40–69%	Moderate
$< 40\%$	Severe

In general, a PEFR <200 L/min indicates severe airways obstruction, and hypercapnia does not occur until the PEFR falls below 25% of predicted (8).

LIMITATION: The PEFR is effort-dependent, and accurate measurements requires a maximum inspiratory effort (to total lung capacity) followed by a maximum expiratory effort (to residual lung volume). Patients with acute exacerbations of asthma and COPD are typically unable to perform these maneuvers because of respiratory distress, and monitoring PEFR has not been popular in the acute care setting (9). However, after the respiratory distress subsides, periodic measurements of the PEFR can be used to monitor the course of the illness.

Intrinsic PEEP

One of the consequences of severe airflow obstruction is the inability to fully exhale the volume of air that is inhaled. The retained volume causes hyperinflation of the lungs and a positive pressure in the alveoli at the end of expiration. This positive end-expiratory pressure (PEEP), which is called *intrinsic PEEP*, is a reflection of the severity of airflow obstruction, and it can be monitored during mechanical ventilation. Intrinsic PEEP is described in more detail later in the chapter.

Aerosol Drug Delivery

Aerosolized bronchodilators are the backbone of management for obstructive airways disease, and there are three devices used to generate these aerosols: the *nebulizer*, the *metered-dose inhaler*, and the *dry powder inhaler*. There are three types of nebulizers (jet, mesh, and ultrasonic nebulizers), but the description here is limited to the jet nebulizer (the traditional device). The design of a jet nebulizer and a metered dose inhaler is shown in Figure 23.1. (Note: Dry powder inhalers are rarely used for life-threatening airways obstruction, and are not described here.)

Jet Nebulizer

The pneumatic or jet nebulizer uses the same principle as an air-entrainment mask. A high-pressure gas source (e.g., 50 psi from a wall outlet) is passed through a narrow opening in the nebulizer, creating a high-velocity (jet) stream of gas that is passed over the opening of a narrow tube submerged in a drug solution. The gas jet draws the drug solution up the tube (by creating viscous drag) and then pulverizes the solution to create an aerosol spray that is inhaled by the patient. Small-volume jet nebulizers have a reservoir volume of 3–6 mL, and can completely aerosolize the reservoir volume in less than 10 minutes (10). Large-volume nebulizers have a reservoir volume >200 mL, and are used for continuous aerosol therapy (see later).

LUNG DEPOSITION: Although small-volume nebulizers can completely aerosolize a drug solution, only a fraction of the drug aerosol reaches the lungs. This is demonstrated in Table 23.1, which shows the distribution of aerosolized albuterol with different aerosol generator systems (11). When the nebulizer is used, most of the drug aerosol impacts on the delivery apparatus or is exhaled, and only 12% of the intended dose reaches the lungs. Inefficient drug delivery is characteristic of aerosol drug therapy, and is not specific for the jet nebulizer.

TABLE 23.1 Distribution of Aerosolized Albuterol by Delivery System			
Site of Deposition	Nebulizer (2.5 mg)	MDI (200 µg)	MDI + Spacer (200 µg)
Exhaled Gas	20%	1%	1%
Apparatus	66%	10%	78%
Oropharynx	2%	80%	1%
Lungs	12%	9%	20%

From Reference 11. MDI = metered dose inhaler. The MDI dose of 200 µg is equivalent to 2 puffs.

Metered Dose Inhaler

A metered dose inhaler (MDI) operates like a canister of hair spray. The MDI has a pressurized canister that contains a drug solution with a boiling point below room temperature. When the canister is squeezed between the thumb and fingers, a valve opens that releases a fixed volume of the drug solution. The liquid immediately vaporizes when it emerges from the canister, and a liquid propellant in the solution creates a high velocity spray.

LUNG DEPOSITION: The spray generated by an MDI has a velocity in excess of 30 meters per second (over 60 miles per hour) (12), and when delivered directly into the mouth, most of the spray impacts on the posterior wall of the oropharynx (called “inertial impaction”), and is not inhaled. The data in Table 23.1 shows that 80% of the aerosolized drug is deposited in the oropharynx, and only 9% reaches the lungs. Drug deposition in the lungs can be increased by adding a holding chamber or “spacer” between the MDI and the mouth, which reduces the spray velocity. This is confirmed by the data in Table 23.1, which shows a more than two-fold increase in the inhaled drug spray when a spacer was used with the MDI. *Spacers are recommended for all bronchodilator treatments that use MDIs (10).*

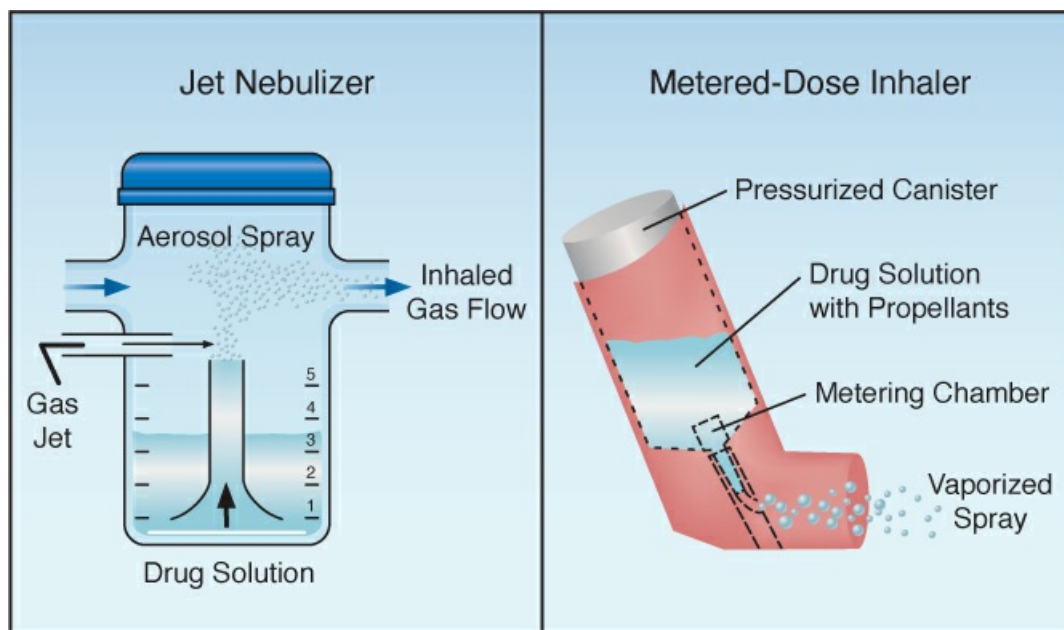


FIGURE 23.1 Aerosol generators used to deliver bronchodilator drugs by inhalation. See text for explanation.

Nebulizer vs. Metered-Dose Inhaler

Bronchodilator treatments with nebulizers and MDIs produce equivalent results despite a marked difference in drug dosage. This is demonstrated in [Figure 23.2](#) which shows a comparison of the bronchodilator responses to three treatments with albuterol given by nebulizer or MDI (with spacer) in patients with acute exacerbation of asthma ([13](#)). Note the similarity in bronchodilator responses with nebulizers and MDIs, despite a 7-fold difference between the nebulizer dose (2.5 mg) and the MDI dose (0.36 mg, or 4 puffs). Adjusting for the inhaled dose (from the percentages in [Table 23.1](#)) yields an albuterol dose of 12% of 2.5 mg (300 μ g) for the nebulizer, and 20% of 0.36 mg (72 μ g) for the MDI, which is a 4-fold difference in the inhaled drug dosage. This dose-response discrepancy is unexplained.

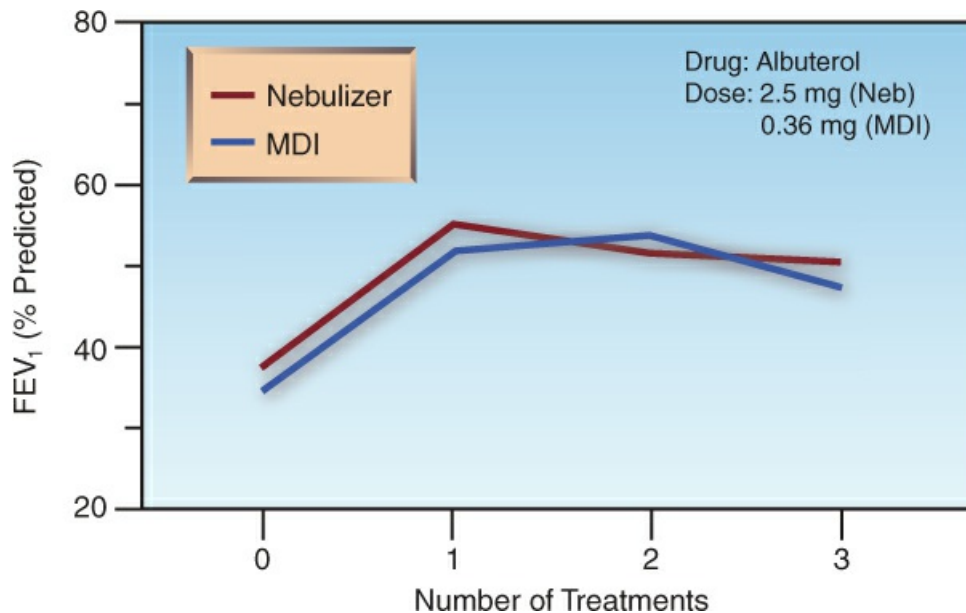


FIGURE 23.2 Comparison of the bronchodilator response to albuterol given by nebulizer or metered-dose inhaler (MDI) in patients with acute exacerbation of asthma. Three treatments were given at 30-minute intervals, using doses indicated in the upper right portion of the graph. FEV₁ = forced expiratory volume in one second. Data from Reference 16.

Mechanical Ventilation

During mechanical ventilation, aerosolized albuterol is delivered through the inspiratory arm of the ventilator tubing using a nebulizer or MDI with spacer. Unfortunately, aerosol deposition in the lungs is impaired further during mechanical ventilation, thanks to condensation on the ventilator tubing and endotracheal tube (14). Aerosol deposition in the lungs can be enhanced by decreasing the inspiratory flow rate and increasing the duration of inspiration (15).

Despite less aerosol deposition in the lungs during mechanical ventilation, the dose of aerosolized albuterol is the same for spontaneous breathing patients and during mechanical ventilation (see later for the doses). And once again, there is no difference in bronchodilator responses with nebulizers or MDIs (16).

Which Method is Preferred?

MDIs (with spacers) are generally preferred because of the lower drug dose, which minimizes side effects. Some people have difficulty coordinating their breathing with drug release from the MDI, but the introduction of breath-actuated MDIs should eliminate this problem. Nebulizers are preferred for acute exacerbations of asthma and COPD, but a switch to MDIs is initiated when the condition stabilizes.

ACUTE EXACERBATION OF ASTHMA

The flow diagram in Figure 23.3 summarizes the early management of adults with acute exacerbation of asthma, and is from the National Asthma Education Program (1). This protocol is based on objective measures of airway obstruction (e.g., peak expiratory flow rate), but clinical measures of disease severity (e.g., tachycardia, use of accessory muscles, etc.) are also

suitable (17,18). Note that a normal arterial PCO_2 ($\text{PaCO}_2 = 42$ mm Hg) despite bronchodilator treatments warrants admission to the ICU. This is explained by the minute ventilation, which is increased (often doubled) in acute asthma, and would normally result in a decrease in PaCO_2 . Therefore, a normal PaCO_2 in the face of a high minute ventilation represents CO_2 retention, and is a sign of severe airway obstruction.

The following is a description of the recommended drugs and dosing regimens for acute exacerbations of asthma. These are summarized in Table 23.2

Short-Acting β_2 -Agonists

The favored bronchodilators are drugs that stimulate β_2 receptors in bronchial smooth muscle, and aerosol delivery is preferred because it is more effective than oral (19) or intravenous (20) drug delivery, and has fewer side effects. Short-acting β_2 -agonists (SABAs) are preferred for acute exacerbations of asthma, while long-acting β -agents are favored for maintenance therapy.

Albuterol is the most widely used SABA for acute exacerbations of asthma (3). Aerosolized albuterol has a rapid onset of action (less than 5 minutes), with a peak effect at 15–60 minutes, and a duration of 2–5 hours (21). *Levalbuterol* is the R-enantiomer of albuterol (a more active form of the drug) that is equally effective at half the dose. However, clinical studies have not shown an advantage with levalbuterol over albuterol (1,17).

Aerosol Regimens

The following regimens of aerosolized albuterol are recommended for acute exacerbations of asthma (3,17,18):

- . The initial treatment involves up to three 20-minute treatments using 2.5 to 5 mg albuterol by nebulizer or 4–8 puffs (90 μg per puff) by MDI with a spacer. Nebulizer delivery is favored for patients with severe airflow obstruction (3), although there is no evidence that nebulizers produce better bronchodilator responses than MDIs (22).
- . If further therapy is needed, albuterol can be given hourly for up to 3 hours, or can be given by continuous nebulization using doses of 5–15 mg/hr. Continuous aerosol therapy is popular, and may be more effective than intermittent aerosol therapy in patients with severe airflow obstruction (23).
- . For patients who are admitted to the hospital, albuterol (2.5–5 mg by nebulizer or 4–8 puffs by MDI) is given every 4–6 hours for the duration of the hospital stay.

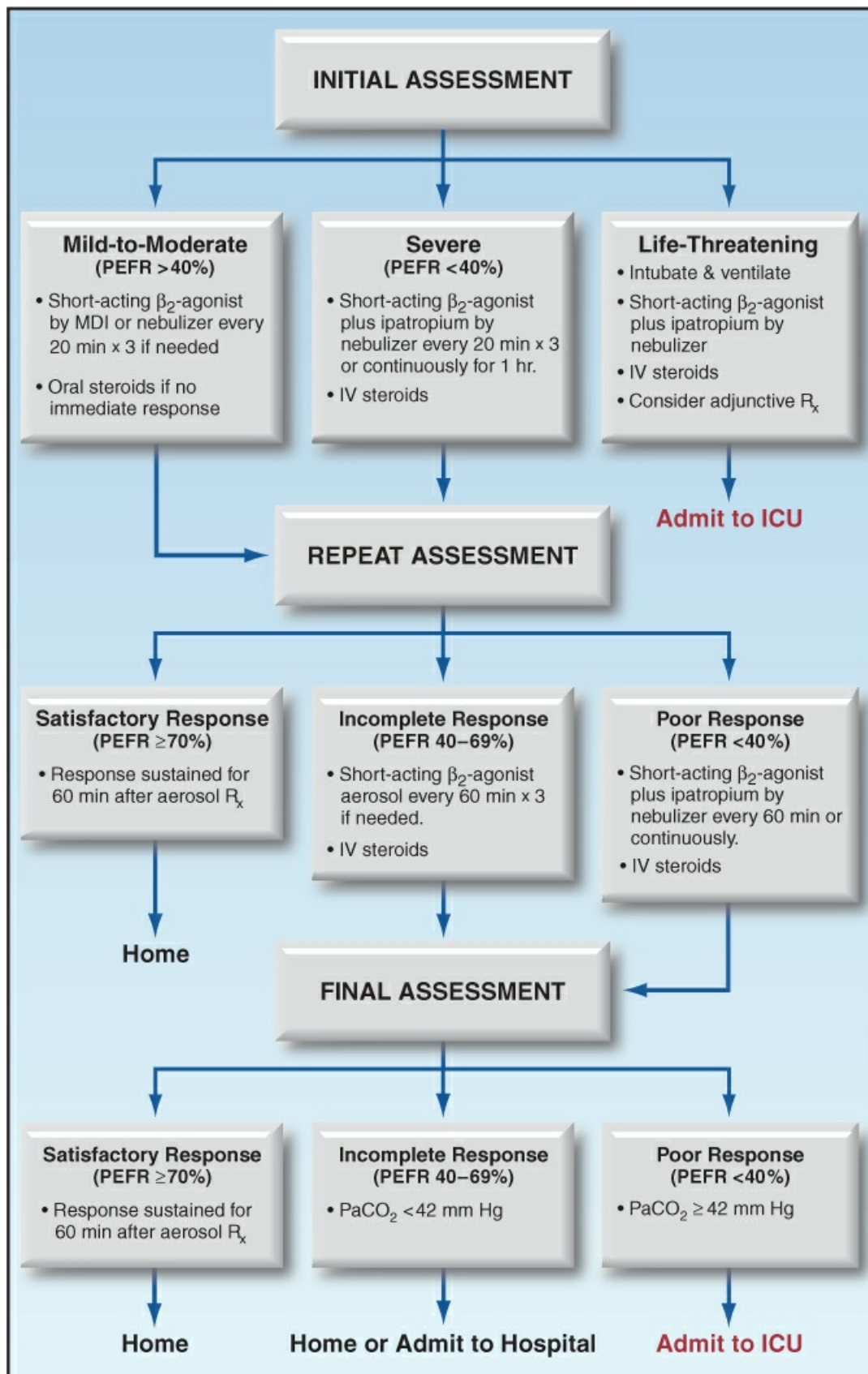


FIGURE 23.3 Flow diagram showing the early management of acute exacerbations of asthma, as recommended in the National Asthma Education Program (3). PEFr = peak expiratory flow rate.

Parenteral Therapy

For the rare asthmatic patient who does not tolerate bronchodilator aerosols (usually because of excessive coughing) parenteral therapy can be given using subcutaneous *epinephrine* (0.3 to 0.5 mg every 20 minutes for 3 doses) or subcutaneous *terbutaline* (0.25 mg every 20 minutes for 3 doses) (3).

Side Effects

Aggressive aerosol therapy with SABAs can produce a number of side effects, including tachycardia, tremors, hyperglycemia, hypokalemia, hypomagnesemia, hypophosphatemia, and elevated plasma lactate levels (24–26). The decrease in serum potassium is the result of a β -receptor-mediated shift of potassium into cells. The source of the elevated lactate levels is likely to be an increase in the metabolic rate, and not anaerobic metabolism (26).

TABLE 23.2 Drug Regimens for Acute Exacerbations of Asthma	
Drug	Dosing Regimens
Albuterol	Dosing: 2.5–5 mg (or 4–8 puffs) every 20 min for 3 doses, then if needed, continue hourly doses, or start continuous inhalation at 5–15 mg/hr, for up to 3 hrs. Maintenance dose is 2.5 mg (or 2 puffs) every 4–6 hrs. Comment: Aggressive dosing can elevate lactate levels.
Levalbuterol	Dosing: 1.25–2.5 mg (or 4–8 puffs) every 20 min for 3 doses, then if needed, continue hourly doses for up to 3 hrs. Maintenance dose is 1.25 mg (or 2 puffs) every 4–6 hrs. Comment: No proven benefit over albuterol.
Ipratropium	Dosing: 0.5 mg every 20 min for 3 doses, then as needed. Can be added (0.5 mg/hr) to continuous inhalation of albuterol. Comment: Use only if initial response to albuterol is not satisfactory, but do not continue use after admission to hospital.
Corticosteroids	Dosing: 40–80 mg of prednisone (PO) or methylprednisolone (IV) initially, then continue daily, in 1 or 2 divided doses, for 7–10 days. Comment: Benefit is not immediate, and takes up to 12 hrs to develop. Taper not necessary after a 10-day course of therapy.
Magnesium	Dosing: 2 grams of MgSO ₄ IV over 15–30 min. Comment: Use only if initial response to albuterol is not satisfactory.

Anticholinergic Agents

Anticholinergic agents offer only marginal benefits in acute asthma, and are used in combination with a SABAs for severe exacerbations, and only for the first 3–4 hours of therapy (3,27).

The only anticholinergic agent approved for use in asthma is *ipratropium bromide*, a derivative of atropine that blocks muscarinic receptors in the airways. The dose in acute asthma is 0.5 mg (which can be mixed with albuterol in the nebulizer) every 20 minutes for 3 doses, then

as needed, or 8 puffs (18 µg per puff) by MDI every 20 min as needed, for up to 3 hours (3). Systemic absorption is minimal, and there is little risk of anticholinergic side effects (e.g., tachycardia, dry mouth, blurred vision, urinary retention). Ipratropium has no proven benefit beyond the first few hours of management, and it *should be discontinued in patients who are admitted for continued asthma management* (3).

Corticosteroids

Corticosteroids are considered a staple in the management of both acute and chronic asthma because of their anti-inflammatory effects. Clinical studies have shown that corticosteroids can hasten the resolution of acute asthma and reduce the risk of relapses (28), although not all studies show a benefit from corticosteroids in acute asthma (29,30). The following observations about steroid therapy in asthma deserve mention:

- . There is no difference in efficacy between oral and intravenous steroids (28,31).
- . The beneficial effects of steroids are often not apparent until 12 hours after therapy is started (31), so steroid therapy will not influence the clinical course of asthma in the emergency department.
- . There is no apparent dose-response curve for steroids (31), and no evidence that doses above 100 mg of prednisone daily (or equivalent doses of other steroids) provide added benefit in acute asthma (28).
- . A 10-day course of steroids can be stopped abruptly without a tapering dose (28,32).

Regimen in Acute Asthma

The National Asthma Education Program includes the following recommendations for corticosteroid therapy in acute exacerbations of asthma (3).

- . Steroids are recommended for all patients who do not show a satisfactory response after one or two bronchodilator treatments.
- . Oral steroids are recommended for patients who can tolerate oral medications.
- . The recommended dose is 40–80 mg daily of *prednisone* (for oral therapy) or *methylprednisolone* (for intravenous therapy) in one or two divided doses, which is continued until there is evidence of satisfactory resolution.
- . Inhaled corticosteroids can be started at any time during treatment of an acute exacerbation of asthma, and are continued after systemic steroids are discontinued, to reduce the risk of relapses.

Mechanism of Action?

The beneficial effects of steroids in acute asthma are attributed to their anti-inflammatory actions. However, the comparative features of corticosteroids in Table 23.3 indicates that *dexamethasone is the most potent anti-inflammatory corticosteroid* (33), yet it is not recommended for the treatment of asthma. This raises questions about the actions of steroids in asthma (as well as the numerous other inflammatory conditions where steroids other than dexamethasone are recommended).

TABLE 23.3**Comparison of the Therapeutic Steroids**

Corticosteroid	Equivalent Dose (mg)	Relative Anti-inflammatory Activity	Relative Mineralocorticoid Activity
Hydrocortisone	20	1	20
Prednisone	5	3.50	1
Methylprednisolone	4	5	0.5
Dexamethasone	0.75	30–40	0

Adjunctive Measures

The following measures can be helpful when acute asthma does not resolve after initial bronchodilator therapy.

Magnesium

Magnesium has mild bronchodilator effects (possibly as a result of calcium channel blockade), and intravenous magnesium (2 grams MgSO_4 infused over 15–30 minutes) can hasten the resolution of acute asthma when the response to albuterol, ipratropium, and steroids is not satisfactory (34). Clinical studies of inhaled magnesium (150 mg MgSO_4) have produced inconsistent results (35), and the inhaled route is not currently advised.

Ketamine

Sedation can be beneficial by reducing the respiratory rate, which will reduce the tendency for air trapping and hyperinflation of the lungs (see later for a description of air trapping). Sedation with ketamine also provides some bronchodilation, and there are several reports of ketamine facilitating the resolution of acute asthma in patients who were refractory to initial bronchodilatory therapy (36). The effective dose is 0.1–0.2 mg/kg IV as a loading dose, followed by a continuous infusion at 0.15–0.25 mg/kg/hr (for 1 to 5 hours) (36). Reported side effects include dysphoria, hallucinations, and increased secretions.

Oxygen Therapy

Hypoxemia is common in severe exacerbations of asthma, and supplemental O_2 should be used to maintain the arterial O_2 saturation (SaO_2) at $\geq 90\%$. Conventional O_2 delivery systems can provide O_2 at rates up to 15 L/min, but inspiratory flow rates in acute asthma can exceed 60 L/min, so a high-flow nasal O_2 system (which delivers heated and humidified O_2 at rates of 40–60 L/min) may be necessary to correct hypoxemia. (High-flow nasal O_2 systems are described in Chapter 25.)

Noninvasive Ventilation

The most severe cases of acute asthma are accompanied by hypercapnia, and these cases may require a trial of noninvasive ventilation with “bilevel positive airway pressure” (BiPAP), which is described in Chapter 26. Failure of BiPAP to lower the arterial PCO_2 is often an indication for

intubation and mechanical ventilation.

ACUTE EXACERBATION OF COPD

Acute exacerbations of COPD are most often the result of a respiratory infection, a covert pulmonary embolus, or left-sided heart failure (5); in some cases, a precipitating event is never identified. Exacerbations that warrant ICU admission are typically those with hypoxemic and hypercapnic respiratory failure, with progressive hypercapnia that may depress mentation.

The management of COPD exacerbations, other than treating any underlying cause, is centered on bronchodilators and corticosteroids, very much like the management of acute asthma. This is a curious approach, since *COPD is characterized by a lack of bronchodilator responsiveness on pulmonary function tests* (5).

Bronchodilator Therapy

The same bronchodilators used in acute asthma are recommended for acute exacerbations of COPD, but the dosing regimens differ, as shown in Table 23.4. Ipratropium is used as combination therapy when the response to short-acting β_2 -agonists is less than satisfactory, although at least three clinical studies do not support this practice (37).

TABLE 23.4 Drug Regimens for Acute Exacerbations of COPD	
Drug	Dosing Regimens
Albuterol	Dosing: 2.5–5 mg by nebulizer, or 2–8 puffs by MDI with spacer, every 4–6 hrs. Comment: MDI favored over nebulizer because of equivalent bronchodilator response at a much lower dose.
Levalbuterol	Dosing: 1.25–2.5 mg by nebulizer, or 2–8 puffs by MDI with spacer, every 4–6 hrs. Comment: More potent form of albuterol, but has no proven advantage.
Ipratropium	Dosing: 0.5 mg by nebulizer, or 2–8 puffs by MDI with spacer, every 4–6 hrs. Comment: Use as combination therapy only when response to short-acting β_2 -agonists is less than satisfactory.
Corticosteroids	Dosing: 40 mg of prednisone-equivalents daily for 5 days. Comment: Oral therapy is equivalent to intravenous therapy.

Corticosteroids

Corticosteroid therapy is associated with a faster recovery and fewer treatment failures in acute exacerbations of COPD (5). *At least 9 patients must be treated with corticosteroids to avoid one treatment failure* (38), but on the other hand, *a steroid-related adverse event occurs once in every 6 patients treated* (38). However, since an adverse effect of steroids (e.g., hyperglycemia) is likely to be less troublesome than a treatment failure, the risk/benefit ratio is in favor of steroid therapy.

The recommended steroid regimen for COPD exacerbations is 40 mg prednisone (or equivalent doses of other steroids) daily for 5 days (5). The intravenous route offers no

advantage over the oral route (5,38,39).

Antibiotics

Airways infections are common triggers for COPD exacerbations, but only about 50% are caused by treatable bacterial infections (40), and this limits the benefit of antimicrobial therapy.

Indications

The most current clinical practice guidelines (5) recommend antibiotic therapy for the following conditions (other than pneumonias): an increase in sputum production, grossly purulent sputum, or the need for noninvasive or invasive mechanical ventilation. The latter conditions apply to most ICU admissions for exacerbations of COPD, so it is reasonable to assume that *all ICU admissions for acute exacerbations of COPD are candidates for antibiotic therapy*.

Empiric Treatment

Sputum cultures are problematic in COPD because the same organisms are often isolated during stable periods and in acute exacerbations (41). Common isolates include *Hemophilus influenzae* and *Streptococcus pneumoniae* (41), as well as gram-negative enteric organisms (e.g., *Pseudomonas aeruginosa*) in ventilator-dependent patients (42). Empiric coverage for both gram-positive and gram-negative pathogens (e.g., cefepime) seems wise for ICU patients, and the recommended duration of antibiotic coverage is 5–7 days (5).

Oxygen Therapy

In cases of severe COPD with chronic hypercapnia, high concentrations of inhaled O₂ can promote further increases in arterial PCO₂. This was originally attributed to loss of hypoxic ventilatory drive (43), but more recent studies have shown that the oxygen-induced rise in arterial PCO₂ is not accompanied by a decrease in ventilatory drive (44). Oxygen unloading of CO₂ from hemoglobin (i.e., the Haldane Effect) may play a role in this phenomenon. Regardless of the mechanism, it is important to avoid high concentrations of inhaled O₂ in patients with chronic CO₂ retention. The goal of oxygen therapy in COPD exacerbations is an arterial O₂ saturation (SaO₂) of 88–92% (and no higher) (5).

Noninvasive Ventilation

Exacerbations of COPD with troublesome hypercapnia should be managed initially with noninvasive ventilation using “bilevel positive airway pressure” (BiPAP) (45), which is described in Chapter 26. Failure of BiPAP to reduce the arterial PCO₂ by 10 mm Hg after one hour is an indication for intubation and mechanical ventilation (46).

POSITIVE-PRESSURE VENTILATION

Patients admitted to the ICU because of asthma or COPD often receive some form of positive-pressure ventilatory assistance (either noninvasive measures like BiPAP or conventional mechanical ventilation), and the following are some important considerations related to positive

pressure breathing in these patients.

Dynamic Hyperinflation

During spontaneous breathing in normal subjects, the volume of air (or gas) inhaled is completely exhaled before the end of expiration. In this situation, there is no expiratory airflow at the end of expiration, so the pressure in the distal airspaces is equivalent to atmospheric (zero reference) pressure. This is illustrated in the lower pressure-volume loop in [Figure 23.4](#). In patients with severe airways obstruction from asthma or COPD, exhalation is prolonged, and is not completed before the next inhalation. This results in hyperinflation, called *dynamic hyperinflation*, and the trapped gas in the distal airspaces creates a positive end-expiratory pressure (PEEP), which is called *intrinsic PEEP* or *auto-PEEP* (47). This is illustrated by the upper pressure-volume loop on [Figure 23.4](#). In this situation, breathing occurs on a flatter portion of the pressure volume curve, which means that the respiratory muscles must generate a higher transpulmonary pressure to draw a given volume of air into the lungs. This increases the work of breathing in patients with severe airflow obstruction. (If you want to feel what it is like to breathe in a hyperinflated state, take a deep breath, and try breathing from there.)

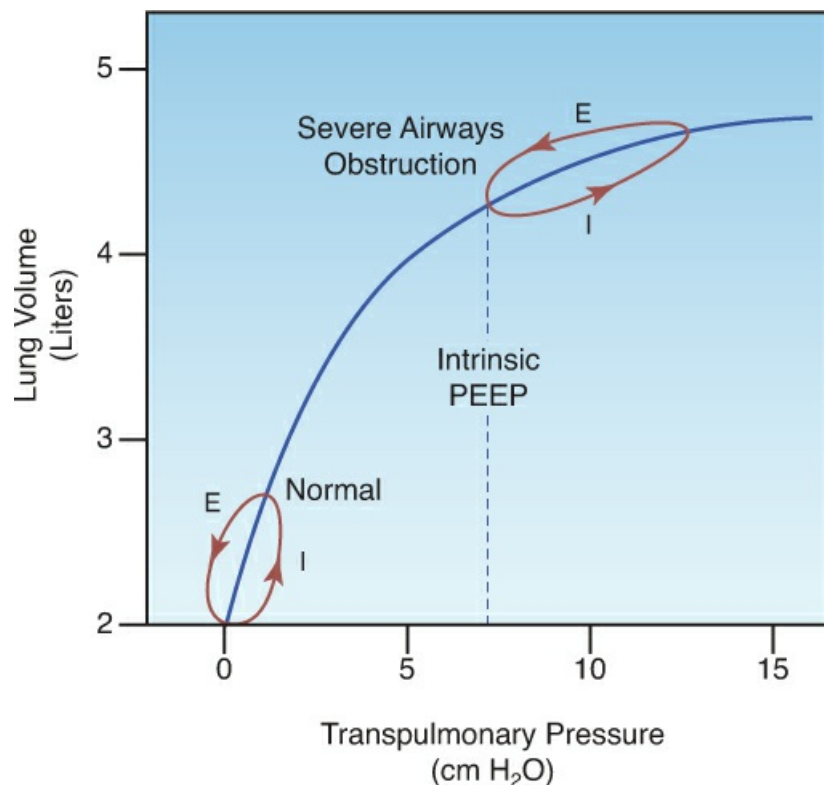


FIGURE 23.4 Pressure-volume curves showing the effects of severe airways obstruction on lung volumes and transpulmonary pressures. The hysteresis loops show the pressure and volume changes during inspiration (I) and expiration (E) for a single breath, See text for further explanation.

Positive Pressure Ventilation

The positive intrathoracic pressures created by positive-pressure ventilation will be increased in the presence of dynamic hyperinflation and intrinsic PEEP (because you are starting at a higher

pressure). In addition, positive-pressure ventilation can aggravate the hyperinflation (by delivering high inflation volumes or rapid rates) (48), and this will produce further increases in intrathoracic pressure. The elevated intrathoracic pressures can have two adverse consequences: (a) the increase in peak intrathoracic pressure promotes *barotrauma* (e.g., pneumothorax), and (b) the increase in mean intrathoracic pressure can impede venous return to the heart. The following measures can help to reduce the risk of these adverse consequences.

Monitoring

Dynamic hyperinflation can be detected by monitoring the flow waveforms during mechanical ventilation, as illustrated in Figure 23.5. The normal flow waveforms in the upper panel show that the expiratory flow ceases before the next lung inflation, while the flow waveforms in the lower panel show that expiratory flow is continuing when the next lung inflation is delivered. *The presence of expiratory flow at the end of expiration is evidence of dynamic hyperinflation.*

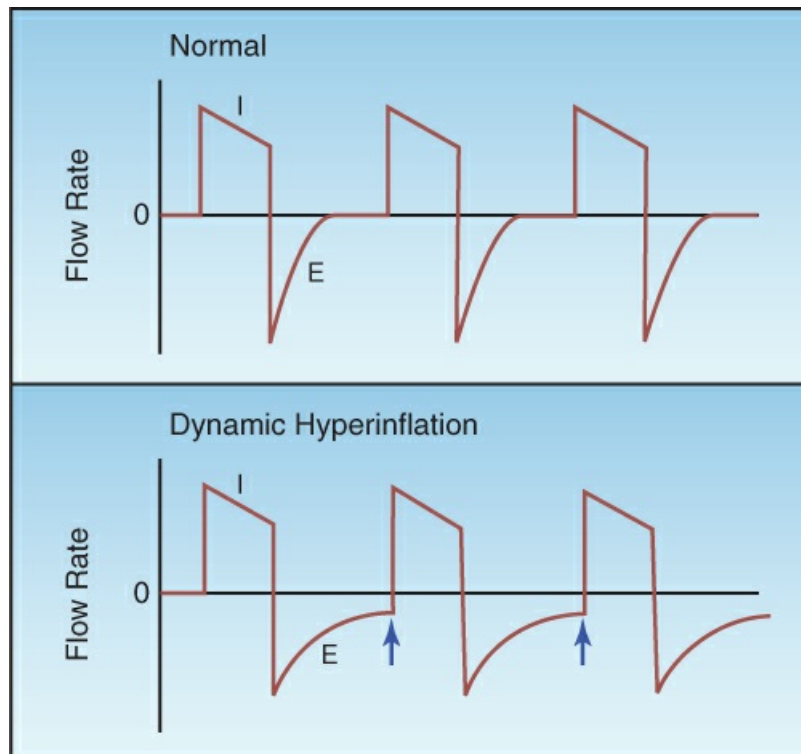


FIGURE 23.5 Flow waveforms during positive pressure mechanical ventilation. The waveforms in the lower panel show continuing expiratory flow at the end of expiration (indicated by the arrows), which indicates the presence of dynamic hyperinflation and intrinsic PEEP. I = inspiration, E = expiration.

INTRINSIC PEEP: When there is evidence of dynamic hyperinflation on the flow waveforms, the severity of the problem can be evaluated by monitoring the level of intrinsic PEEP. (The measurement of intrinsic PEEP is described in Chapter 28.) *The intrinsic PEEP level is a reflection of the severity of airways obstruction in patients with asthma and COPD.*

Ventilator Strategies

The following adjustments are designed to limit dynamic hyperinflation (and the adverse consequences of high intrathoracic pressures) during mechanical ventilation:

- . Avoid inflation volumes that exceed 6–8 mL/kg.
- . Maximize the time allowed for exhalation by: (a) slowing a rapid breathing rate with sedation, if possible, and (b) increasing the inspiratory flow rate, if needed, so that lung inflation accounts for only one-third of the respiratory cycle.

The management of hyperinflation and intrinsic PEEP during mechanical ventilation is described further in [Chapter 28](#).

A FINAL WORD

Keeping it Simple

The management of patients admitted to the ICU because of severe exacerbations of asthma or COPD can be summarized as follows:

- . Give bronchodilators and corticosteroids to all patients, plus antibiotics for patients with severe COPD.
- . If the condition is severe or progresses despite the above measures, use noninvasive ventilation if possible, and conventional mechanical ventilation if necessary.

That's about it.

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Acute Respiratory Distress Syndrome

Physicians think they do a lot for a patient when they give his disease a name.

Immanuel Kant

The disease entity described in this chapter has had several names over the years, including shock lung, Da Nang lung (from the Vietnam war), stiff-lung, leaky capillary pulmonary edema, noncardiogenic pulmonary edema, acute lung injury, adult respiratory distress syndrome, and most recently, *acute respiratory distress syndrome*, or *ARDS*. However, none of these names provides any useful information about the disease, which is a diffuse inflammatory injury of the lungs, and one of the leading causes of acute respiratory failure in modern times (1–3).

PATHOGENESIS

The first clinical report of ARDS appeared in 1967 (4), and included 12 patients with refractory hypoxemia and bilateral, diffuse infiltrates on chest x-ray. Seven patients died, and autopsy findings revealed dense infiltration of the lungs with an inflammatory exudate. There was no evidence of infection, which indicated that ARDS is an acute inflammatory lung injury.

Inflammatory Injury

The lung consolidation in ARDS originates with the activation of circulating neutrophils (5). This leads to neutrophil sequestration in the pulmonary microcirculation, where the neutrophils attach to the vascular endothelium and move between endothelial cells and into the lung parenchyma. The neutrophils then degranulate to release the contents of their cytoplasmic granules (i.e., proteolytic enzymes and toxic oxygen metabolites). Subsequent damage to the capillary walls then leads to exudation of protein-rich fluid, erythrocytes, and platelets into the lungs. The cellular and proteinaceous exudate eventually fills and obliterates the distal airspaces, as shown in [Figure 24.1](#). The inflammatory exudate contains fibrin, which can accumulate and produce irreversible pulmonary fibrosis if the inflammation does not resolve within a few weeks.

Predisposing Conditions

ARDS is not a primary disorder, but is a consequence of a variety of infectious and noninfectious conditions. The typical conditions that predispose to ARDS are listed in [Table 24.1 \(1\)](#). The most frequent offenders are pneumonia, extrapulmonary sepsis, septic shock, aspiration of gastric contents, blood transfusions, noncardiogenic shock, and multisystem trauma (1). The shared feature among all these conditions is the activation of neutrophils in the inflammatory response (see [Figure 17.1](#)), which is the principal inciting event in ARDS. About 10% of cases of ARDS have no identifiable source (1).

COVID-19 and ARDS

The pneumonia caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) is generally classified as ARDS when it causes hypoxemic respiratory failure, and at the height of the coronavirus (COVID-19) pandemic, ARDS was reported in one of every three patients hospitalized with COVID-19 (6). However, severe hypoxemia from COVID-19 is often associated with little or no change in lung compliance (distensibility) (7), which is not a characteristic finding in severe ARDS. Therefore, COVID-19 is a unique type of ARDS, which is relevant because many of the recent studies of ARDS have been in patients with COVID-19, and the results may not apply to other cases of ARDS.

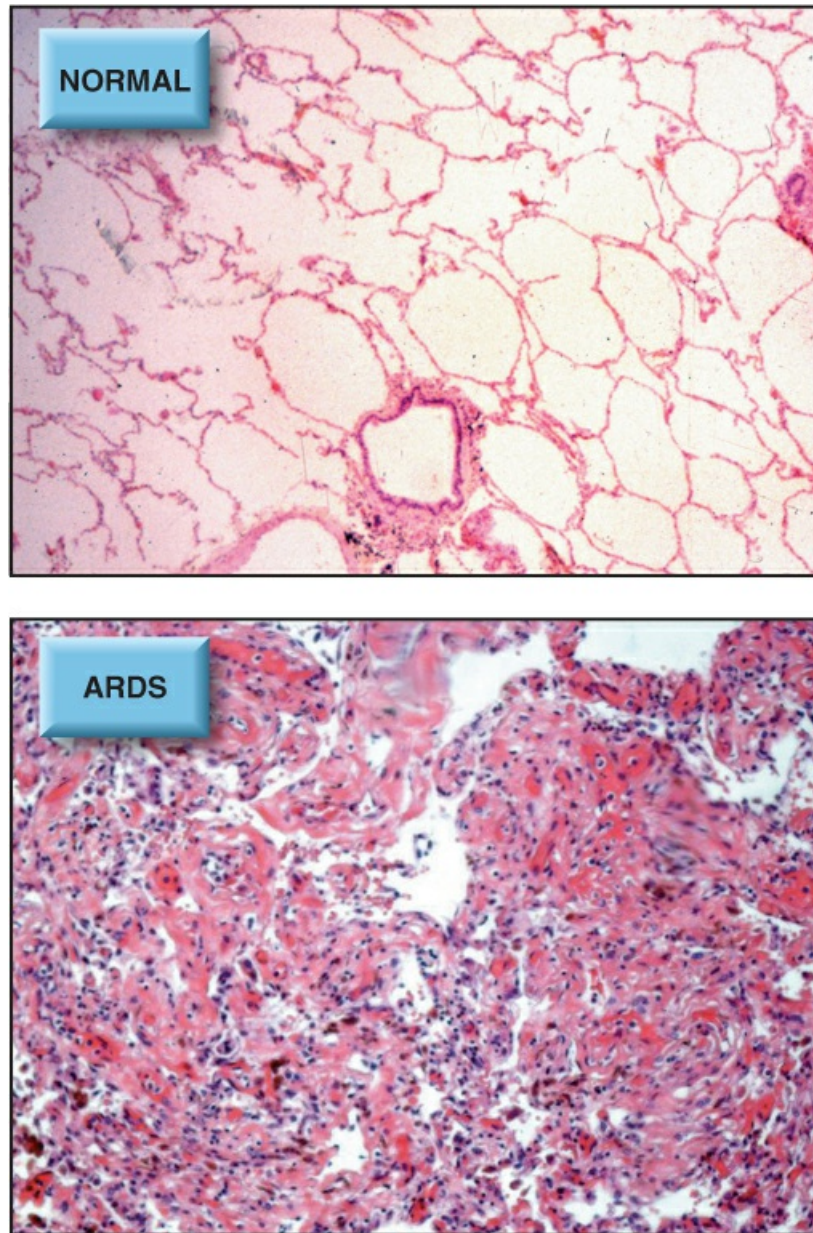


FIGURE 24.1 Microscopic appearance of a normal lung, which shows an abundance of distal airspaces separated by thin segments of tissue that contain the pulmonary capillaries, and a lung in the advanced stages of ARDS, which shows a dense infiltration of leukocytes and proteinaceous material that fills and obliterates the distal airspaces. Photomicrograph of ARDS courtesy of Martha L. Warnock, MD. Image digitally retouched.

TABLE 24.1 Predisposing Conditions for ARDS	
Infection-Related	Noninfectious
Pneumonia (60%) Extrapulmonary Sepsis (16%) Septic Shock (8%)	Gastric Aspiration (14%) Blood Transfusions (4%) Multisystem Trauma (4%) Pulmonary Contusion (6%) Inhalation Injury (2%)

	Drug Overdose (2%)
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From Reference 1. The prevalence of each condition is in parentheses. The total prevalence exceeds 100% because some cases had more than one predisposing condition.

THE DIAGNOSIS OF ARDS

Clinical Features

The clinical criteria for the diagnosis of ARDS are listed in [Table 24.2](#). The principal criteria include the acute onset of hypoxemic respiratory failure (as defined by a $\text{PaO}_2/\text{FIO}_2$ ratio ≤ 300 mm Hg with PEEP = 5 cm H_2O), bilateral infiltrates on the chest x-ray, the presence of a predisposing condition for ARDS, and no evidence of left heart failure or fluid overload (8). The earliest signs of ARDS are progressive dyspnea and hypoxemia. The chest x-ray can be unrevealing in the early hours after symptom onset, but bilateral pulmonary infiltrates begin to appear within 24 hours. The respiratory failure from ARDS often warrants admission to the ICU, and 80% of these patients require intubation and mechanical ventilation, typically in the first 48 hours after presentation (1).

Severity of Illness

ARDS has a severity-of-illness rating based on the $\text{PaO}_2/\text{FIO}_2$ ratio: i.e., a ratio of 201–300 mm Hg is mild illness, 100–200 mm Hg is moderate illness, and <100 mm Hg is severe illness (1). The in-hospital mortality rate is about 35% for mild ARDS, 40% for moderate ARDS, and 45% for severe ARDS (1).

TABLE 24.2	Diagnostic Criteria for ARDS
1. Acute onset (within 7 days). 2. Bilateral alveolar infiltrates on chest x-ray. 3. $\text{PaO}_2/\text{FIO}_2 \leq 300$ mm Hg with PEEP ≥ 5 mm Hg. 4. No evidence of left heart failure or fluid overload. 5. The presence of a predisposing condition.	

From Reference 6. FIO_2 = fractional concentration of inhaled oxygen, PEEP = positive end-expiratory pressure.

Diagnostic Problems

Despite almost 50 years of clinical experience with ARDS, errors in diagnosis (both overdiagnosis and underdiagnosis) are reported in as many as 50% of cases (1,9). The following sections highlight some of the diagnostic pitfalls.

PEEP as a Diagnostic Criterion

The standard definition of ARDS includes a requirement for positive end-expiratory pressure (PEEP), as shown in [Table 24.2](#) (item #3). This has been questioned (2), since the application of PEEP requires intubation and mechanical ventilation, which precludes cases of ARDS that do not require mechanical ventilation.

Chest Radiography

The classic radiographic appearance of ARDS is shown in the top panel of [Figure 24.2](#). The infiltrate has a finely granular or “ground-glass” appearance, and is evenly distributed in all lung fields, with no evidence of a pleural effusion. Unfortunately, the *characteristic features on chest x-ray are present in only 15–25% of cases of ARDS* ([10,11](#)). More localized infiltrates are more common ([10](#)), as demonstrated by the chest x-ray in the lower panel of [Figure 24.2](#), which shows infiltrates that are confined to the lower lung fields, and a probable left-sided pleural effusion. This chest x-ray is from a case of ARDS secondary to gram-negative septicemia, and could be mistaken for cardiogenic pulmonary edema.

The chest x-ray is a major source of error in the diagnosis of ARDS. When compared with computed tomography (CT) of the chest (the most reliable imaging modality for ARDS), chest x-rays have a negative predictive value of only 47% in ARDS ([10](#)), which means a negative chest x-ray has about a 50% chance of excluding the presence of ARDS. This is no better than predicting heads or tails on a coin toss.

Lack of Specificity

Many of the diagnostic criteria for ARDS are shared with other causes of acute respiratory failure (e.g., pneumonia), and this creates a risk for misdiagnoses. This is demonstrated by an autopsy study showing pathological evidence of ARDS in only 50% of the patients who died with a clinical diagnosis of ARDS ([9](#)). The likelihood of identifying ARDS in this study was 50%, similar to the performance of chest x-rays mentioned earlier.

Pitfalls of the Wedge Pressure

When the chest-x-ray shows overlapping features of ARDS and cardiogenic pulmonary edema (like the one in [Figure 24.3](#)), the pulmonary artery “wedge” pressure has been used to distinguish between these two conditions: i.e., a wedge pressure ≤ 18 mm Hg is considered evidence of ARDS ([12](#)). This is problematic because *the wedge pressure is NOT a measure of capillary hydrostatic pressure*, as explained in [Chapter 8](#) (see section titled WEDGE PRESSURE). The left atrial (wedge) pressure is lower than the capillary hydrostatic pressure, which creates the pressure gradient needed for flow in the pulmonary veins. This means that *a normal wedge pressure does not exclude the diagnosis of cardiogenic (hydrostatic) pulmonary edema*. The erroneous interpretation of the wedge pressure has undoubtedly led to the overdiagnosis of ARDS.

Computed Tomography

Computed tomographic (CT) imaging has become a standard practice in the diagnosis of ARDS, and CT images can have a different appearance at different stages of the disease. This is demonstrated in [Figure 24.3](#). The CT image in the top panel is from the early stage of ARDS, and shows dense consolidation in the posterior lung regions, which are the dependent lung regions in the supine position. The CT image in the lower panel is from an advanced stage of ARDS ([13](#)), and shows diffuse, reticulated infiltrates (that may represent pulmonary fibrosis) in all lung fields. The CT appearance in ARDS has important implications for management (see later).

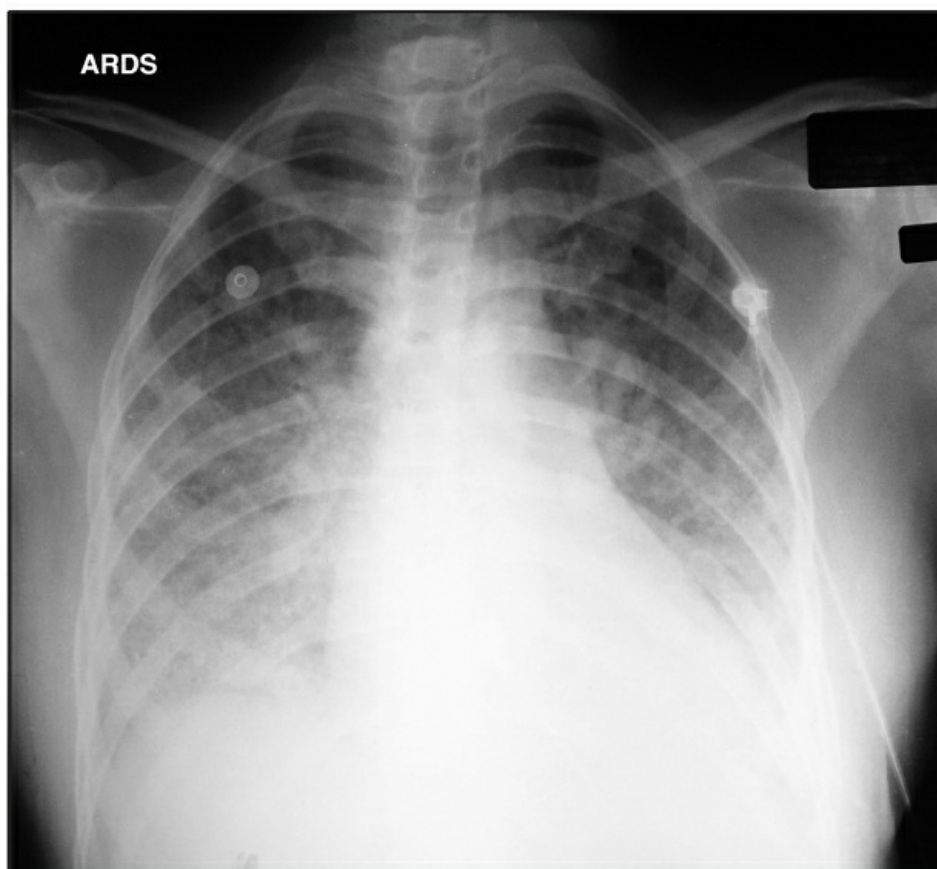


FIGURE 24.2 Variable chest x-ray appearance in ARDS. The top panel shows the classic radiographic appearance, with finely granular infiltrates that are evenly distributed throughout both lungs. The bottom panel (which is from a case of ARDS secondary to gram-negative septicemia) shows bilateral infiltrates that are confined to the lower lung fields, with obliteration of the left hemidiaphragm suggesting a pleural effusion.



FIGURE 24.3 CT images at different stages of ARDS. The image in the top panel is from the

early stage of ARDS, and shows dense consolidation in the posterior lung regions. The image in the lower panel is from an advanced stage of ARDS, and shows reticulated infiltrates in all lung fields, with a pneumo-mediastinum (indicated by the arrow). Images from Reference 7 (top panel) and Reference 13 (lower panel).

NON-VENTILATOR MANAGEMENT

Other than treatment of the predisposing condition (if available), drug therapies aimed at improving survival in ARDS have been marked by failure rather than success. Failed therapies include surfactant (in adults), inhaled nitric oxide, inhaled prostacyclins, statins, corticosteroids (with a few exceptions described later), neuromuscular blockers, β -receptor agonists, interferon β -1a, and granulocyte-colony stimulating factor (14). As a result, the management of ARDS consists primarily of general supportive measures, including mechanical ventilation. This section focuses on support measures other than mechanical ventilation, especially ones that have a proven impact.

Patient Self-Induced Lung Injury (P-SILI)

Avoiding intubation is a priority in ARDS. However, breathing with increased effort (as often occurs in patients with respiratory failure) can be a source of lung injury; a phenomenon known as “patient self-induced lung injury” (P-SILI) (15). This injury is attributed to large negative swings in intrathoracic pressure created by vigorous breathing efforts, as they can promote “negative pressure pulmonary edema” (16). The presumed mechanism is incomplete transmission of the negative intrathoracic pressures across the walls of intrathoracic blood vessels, which increases the transmural pressure that drives fluid out of the pulmonary capillaries (see Figure 24.4). This tendency for edema formation is enhanced further by the “leaky” pulmonary capillaries in ARDS.

The other consequence of increased breathing efforts is an increase in inhaled (tidal) volumes, which can damage the lungs in ARDS by overstretching alveoli in unaffected areas of lung. This is similar to the “volutrauma” in ventilatory-induced lung injury (see later). Of interest, P-SILI can also occur during mechanical ventilation in patients who “overbreathe” the ventilator.

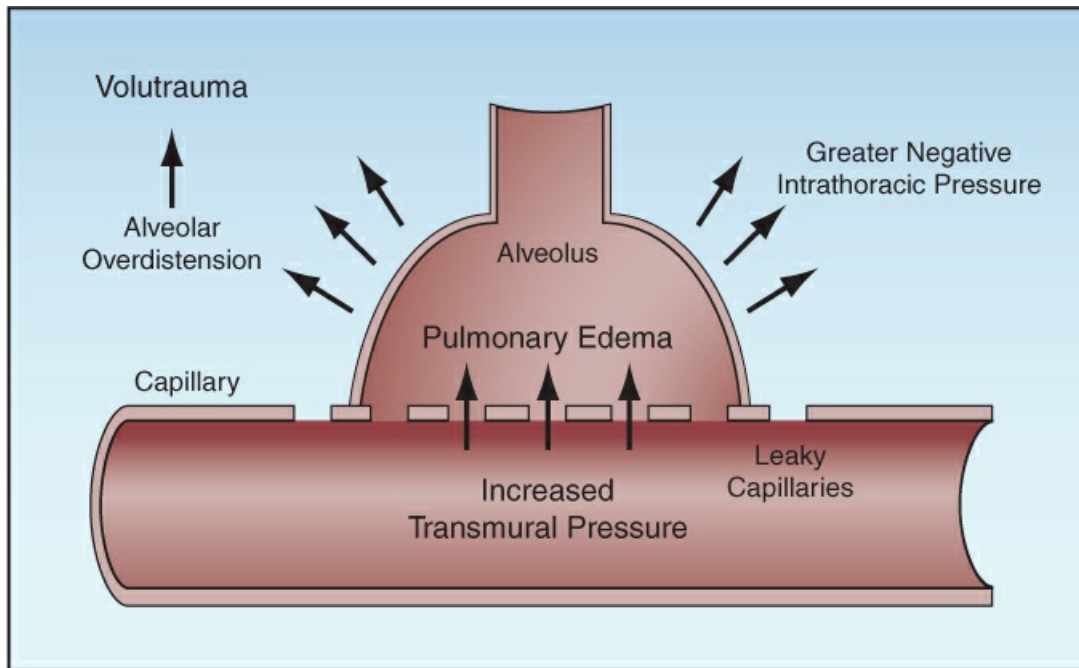


FIGURE 24.4 Schematic illustration of the factors involved in “patient self-induced lung injury” (P-SILI). See text for explanation.

Implications

The risk of P-SILI implies that delays to intubation and mechanical ventilation can have deleterious consequences for patients who continue to have labored breathing. This highlights the importance of measures (e.g., oxygen therapy, sedation) that can relieve the sense of dyspnea and promote comfortable breathing. For patients who continue to breathe with increased efforts, intubation and mechanical ventilation is advised sooner rather than later.

Oxygen Therapy

Providing supplemental O_2 is the first concern in patients with hypoxemic respiratory failure. If conventional O_2 therapy (which delivers O_2 at rates up to 15 L/min) does not correct the hypoxemia, then a trial of high-flow nasal O_2 is warranted.

High-Flow Nasal O_2

High-flow nasal oxygen (HFNO) provides heated and humidified O_2 at rates up to 40–60 L/min through specially-designed nasal prongs, and the circuit has an oxygen regulator that allows independent adjustments of the FI_{O_2} (see [Figure 25.5](#)). Studies of ARDS from COVID-19 ([17](#)) have shown that patients treated with HFNO had fewer intubations than patients treated with conventional O_2 therapy (although there was no difference in mortality rate). The reduction in intubation rate with HFNO is not a consistent finding ([18](#)), but it is now recommended as a *replacement* for conventional O_2 therapy in patients with hypoxemic respiratory failure ([2](#)). The advantage of HFNO over conventional O_2 therapy is the ability to produce comfortable breathing (which is attributed to the high flow rate). This not only promotes patient well-being, it also reduces the risk of patient-induced self injury (P-SILI), and thus could limit the extent of lung

injury.

Don't Forget O₂ Toxicity

The use of high-flow O₂ to delay or prevent intubation and mechanical ventilation often exposes patients to high concentrations of O₂ (i.e., FIO₂ ≥70%) for prolonged periods of time, and this introduces the risk of pulmonary O₂ toxicity. Attention to the risk of pulmonary O₂ toxicity is emphasized by the following statements:

- . Critically ill patients are often deficient in protective antioxidants (e.g., glutathione) (19), and thus are susceptible to O₂ toxicity.
- . High levels of O₂ accelerate the damaging effects of mechanical ventilation (20).
- . The lung injury in pulmonary O₂ toxicity is identical to ARDS, so cases of ARDS that progress while breathing high concentrations of O₂ could be the result of O₂ toxicity.

The above concerns highlight the importance of limiting the concentration of inhaled O₂ whenever possible. (The toxic effects of O₂ are presented in [Chapter 25](#).)

Noninvasive Ventilation

Noninvasive ventilation (NIV), which is described in [Chapter 26](#) is a popular intervention in the early management of ARDS, and is most successful in preventing intubation in patients with mild ARDS (i.e., PaO₂/FIO₂ = 201–300 mm Hg) (21). This is shown in [Figure 24.5](#) (panel on the left). The panel on the right shows that patients who fail NIV and require intubation have a much higher mortality rate. Although this is likely a reflection of the underlying severity of illness, it is also possible that patient self-induced lung injury (P-SILI) plays a role in the adverse outcomes.

P-SILI and NIV Failure

Failure to increase the PaO₂/FIO₂ ratio after one hour of NIV indicates a poor response, and intubation should be considered at this time. Patients who fail NIV typically have increased tidal volumes (i.e., ≥9.5 mL/kg predicted body weight) (22), indicating an increased breathing effort, and suggesting the possible role of P-SILI in NIV failure. As such, intubation should be a serious consideration when the PaO₂/FIO₂ ratio does not improve and the patient continues to have increased breathing efforts.

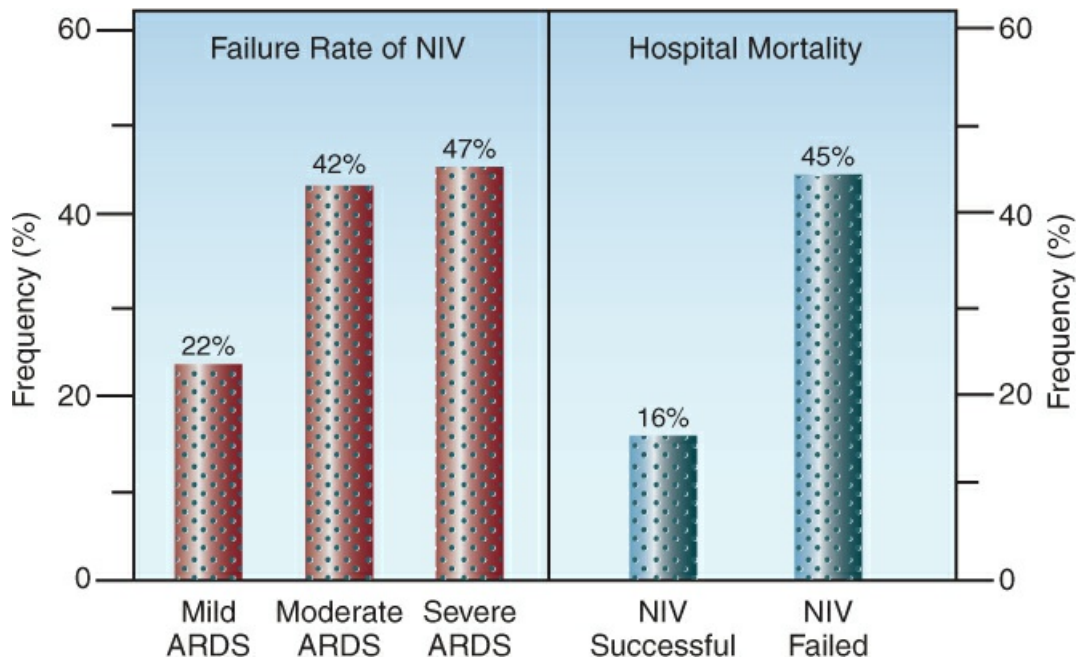


FIGURE 24.5 Observations on the use of noninvasive ventilation (NIV) for the first 48 hours after the diagnosis of ARDS. Panel on the left shows that NIV was most successful in preventing intubation in patients with mild ARDS ($\text{PaO}_2/\text{FIO}_2 = 201\text{--}300$ mm Hg) and the panel on the right shows that failure of NIV has a higher mortality rate. Data from Reference 21. See text for further explanation.

Fluid Management

The lungs are susceptible to fluid accumulation in ARDS because of the leaky pulmonary capillaries, so fluid management should be aimed at avoiding or correcting a positive fluid balance, which can reduce the duration of mechanical ventilation (23). This goal is best served by attention to daily fluid balances, with the use of diuretics to correct fluid excess. The positive intrathoracic pressures during mechanical ventilation will magnify the drop in cardiac output in response to hypovolemia, so it is also important to avoid fluid deficits, which can promote systemic hypoperfusion and aggravate the multiorgan dysfunction that often accompanies ARDS (24).

Corticosteroid Therapy

You may already be aware that steroids enjoy a popularity in clinical medicine that is far greater than deserved, and ARDS provides an example of this phenomenon. There have been innumerable trials of steroid therapy in ARDS since the disease was first described almost 50 years ago, yet the role of steroid therapy in ARDS continues to be debated (13,25), and the recommendations seem to change every 5–10 years. (Simply stated, if a drug effect is not convincing after 50 years, it's time to move on.)

Clinical trials of steroid therapy in ARDS have produced inconsistent results (13,25,26), which is partly attributed to the heterogeneity of the ARDS cases, and to differences in the steroid regimens. The following recommendations are based on evidence of improved outcomes with steroid therapy in at least some clinical trials. This information is summarized in Table 24.3.

COVID-19

Dexamethasone (6 mg IV or PO daily for up to 10 days) is recommended for cases of COVID-19 that require supplemental O₂ or some form of ventilatory assistance (noninvasive or invasive mechanical ventilation) (27). There is no evidence that this steroid regimen impairs clearance of the SARS-CoV-2 virus (28).

Moderate-to-Severe ARDS

The most recent clinical practice guideline includes a weak recommendation for steroid therapy the first 14 days for patients with moderate-to-severe ARDS (PaO₂/FIO₂ ratio <200 mm Hg with PEEP = 10 cm H₂O) (29). There are two steroid regimens (see Table 24.3). Both use intravenous infusions of methylprednisolone. One regimen is initiated in the early stage of disease (i.e., first 72 hours), and uses a methylprednisolone dose of 1 mg/kg/day (ideal body weight). The other regimen is used for ARDS that persists for 7–14 days, and uses a methylprednisolone dose of 2 mg/kg/day. (The latter regimen is aimed at the fibrinoproliferative phase of ARDS, which begins 7–14 days after the onset of illness, and culminates in irreversible pulmonary fibrosis) (30). Both steroid regimens are continued for 14 days, with a slow taper over this time.

Adverse Effects

Hyperglycemia is the most frequently reported side effect of steroid therapy for ARDS (25). The most feared complication is an acute myopathy that is typically associated with the combined use of high dose steroids and neuromuscular blocking agents, but this is rare (31).

TABLE 24.3 Recommendations for Corticosteroid Therapy in ARDS	
Condition	Drug Regimen
COVID-19 that requires any of the following: a. Oxygen therapy b. Noninvasive ventilation c. Mechanical ventilation	Dexamethasone: 6 mg (IV or PO) once daily for up to 10 days.
Moderate-to-Severe ARDS in the first 72 hrs after onset.	Methylprednisolone: 1 mg/kg/day (IBW) by continuous infusion, and taper slowly over 14 days. Can eventually switch to PO therapy using once daily dosing.
Moderate-to-Severe ARDS that persists 7–14 days after onset	Same regimen as above, except the methylprednisolone dose is 2 mg/kg/day (IBW).

From References 27,29. IBW = ideal body weight.

Prone Positioning

CT images like the one at the top of Figure 24.3, which shows a dense consolidation in the posterior lung regions (7,32), prompted studies of prone positioning to redirect blood flow to better aerated anterior lung regions.

Physiological Effects

Prone positioning not only increases blood flow in the anterior region of the lungs, it also promotes aeration in posterior lung segments (33). Both effects serve to improve arterial oxygenation and facilitate CO₂ removal. During spontaneous breathing, a switch to the prone position reduces both the respiratory rate and the work of breathing (34), possibly by increasing lung compliance (i.e., distensibility). There is also a decrease in pulmonary artery pressures (35), which helps alleviate right heart strain.

When to Use

Prone positioning was originally used as a rescue maneuver for refractory hypoxemia in mechanically ventilated cases of ARDS, but the COVID-19 pandemic shifted attention to the use of prone positioning in spontaneously breathing patients, where it proved effective in preventing intubation in select groups of patients. The current indications for prone positioning in ARDS are shown in Table 24.4, and are based on the observations summarized below.

- . In spontaneously breathing patients with ARDS, prone positioning can reduce the need for intubation and mechanical ventilation, but only in patients who are receiving high-flow nasal O₂ or noninvasive ventilation (36). Despite fewer intubations in these patients, there is no impact on the mortality rate (36).
- . In mechanically ventilated patients with ARDS, prone positioning has a proven survival benefit in patients with moderate-to-severe disease (i.e., PaO₂/FIO₂ <150 mm Hg with PEEP ≥5 cm H₂O, or an FIO₂ ≥60% with PEEP of 10 cm H₂O) who are ventilated at low tidal volumes (about 6 mL/kg predicted body weight) (37).

CONCERNS: The ability of prone positioning to prevent intubation is derived from studies of COVID-19 ARDS, and the results may not apply to ARDS from other sources (because COVID-19 is a unique form of ARDS, as mentioned earlier). In addition, the survival benefit of prone positioning in mechanically ventilated patients is not a consistent finding (38).

One proposal that deserves attention is *the possibility that the success of prone positioning is dependent on appearance of the CT scan* (7), with a greater likelihood of success when the CT scan shows dense consolidation in the posterior lung regions.

TABLE 24.4 Prone Positioning in ARDS	
Indications	Contraindications [†]
1. To prevent intubation in patients who are breathing spontaneously but require high-flow nasal O₂ or noninvasive ventilation. 2. To promote survival in mechanically ventilated patients with moderate-to-severe ARDS* who are ventilated with a tidal volume of ~6 mL/kg (predicted body weight).	Active hemorrhage Spinal instability Multiple fractures Increased ICP Recent tracheostomy or sternotomy (2 wks)

* PaO₂/FIO₂ <150 mm Hg and FIO₂ ≥60%, with PEEP ≥5 cm H₂O.

[†]Includes only absolute contraindications, from Reference 40. ICP = intracranial pressure.

CONTRAINDICATIONS: The absolute contraindications for prone positioning are listed in [Table 24.4 \(39\)](#). Relative contraindications include circulatory shock, pregnancy, recent life-threatening arrhythmias, and functioning chest tubes (39). The use of vasopressors is not a contraindication if the clinical condition is stable or improving.

How to Use

ICUs usually have a protocol for prone positioning, and the following are some important considerations for this protocol.

DURATION: The duration of prone positioning is an important determinant of success. For spontaneously breathing patients, at least 8 hours of prone positioning daily (in a single session) is recommended (40). For mechanically ventilated patients, at least 16 hours in the prone position each day is advised (2).

EVALUATION: A beneficial response (e.g., increase in the $\text{PaO}_2/\text{FI}\text{O}_2$ ratio, increase in lung compliance) should be seen after the first session of prone positioning; if the improvement persists for 4 hours after the session, then prone positioning is no longer needed (38). If there is no evidence of benefit after a session of prone positioning, then further sessions are not warranted.

COMPLICATIONS: Complications of prone positioning include patient discomfort, vomiting, brachial plexus injury, pressure sores, dislodgement of endotracheal tubes and vascular catheters, and transient arrhythmias. These are generally infrequent (39).

MECHANICAL VENTILATION IN ARDS

ARDS is the most common cause of acute hypoxemic respiratory failure that requires mechanical ventilation (1), so the discovery that *mechanical ventilation can be a source of lung injury that resembles ARDS* has profound implications. This section describes the lung injury related to mechanical ventilation, and the strategy that is designed to limit this injury.

Ventilator-Induced Lung Injury

Positive pressure mechanical ventilation rose to prominence during the polio epidemic in the 1950s, and it did not take long to recognize that positive pressure lung inflations could rupture distal airspaces and produce pneumothoraces. A more complete picture of the lung injury from mechanical ventilation emerged in 1967 as the *respirator lung syndrome* (41), which was described as diffuse inflammatory infiltration in the lungs with hyaline membrane formation. Later studies showed that mechanical ventilation produces stress fractures in the alveolar capillary interface, as shown in [Figure 24.6](#), and this damage promotes inflammation and fluid accumulation in the lungs (42). This condition is now known as “ventilator-induced lung injury” (VILI) (42). Multiple factors are involved in this injury, as described next.

Volutrauma

When the lungs are stiff (as occurs in ARDS), ventilators must generate higher pressures to deliver a given tidal volume, and VILI was initially considered to be a pressure-related injury

(i.e., barotrauma). However, a landmark study in the 1980s showed that VILI is the result of high inflation *volumes*, not high inflation airway pressures (43). Instead of barotrauma, the culprit in VILI is *volutrauma*.

There are two predisposing factors for volutrauma. The first is the high tidal volumes used during mechanical ventilation. When positive pressure ventilation was first introduced, large tidal volumes were adopted to reduce the tendency for atelectasis during controlled ventilation (44). The traditional tidal volumes during mechanical ventilation are 12–15 mL/kg (ideal body weight), which is about twice the normal tidal volume (6–7 mL/kg). The second predisposing factor is the reduced volume capacity of the lungs in ARDS, as demonstrated in the CT scan at the top of [Figure 24.3](#). Thus, tidal volumes that are twice normal are being forced into lungs that are operating at 50% capacity. This volume mismatch is the principal source of alveolar rupture, and disruption of the alveolar-capillary interface.

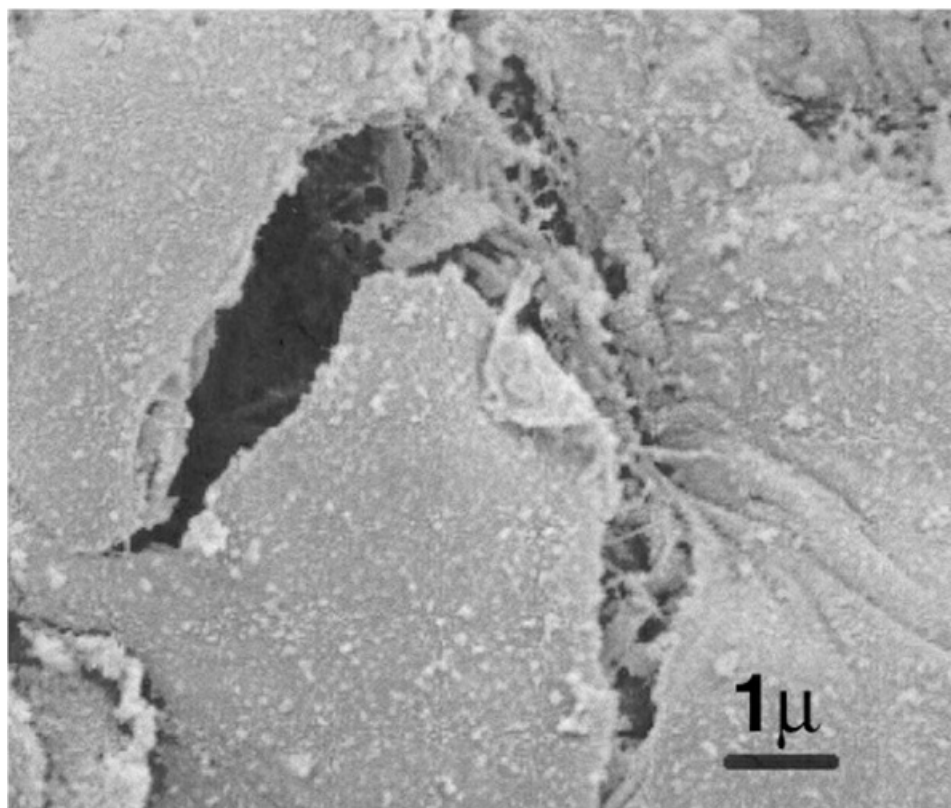
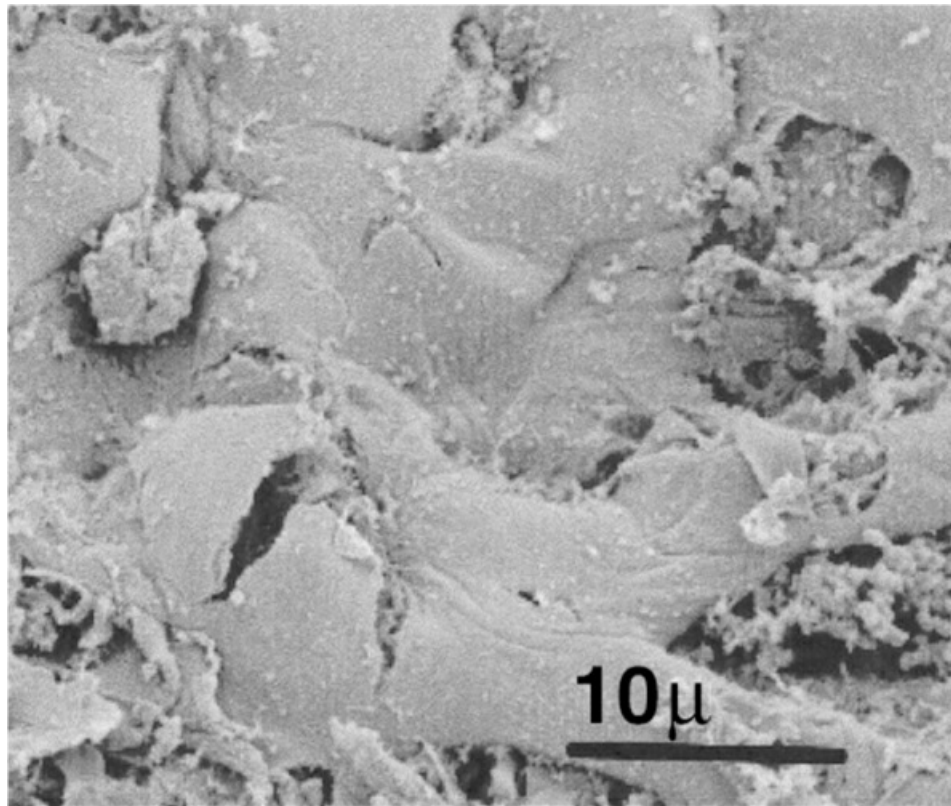


FIGURE 24.6 Electron micrographs of a (postmortem) lung specimen from a patient with ARDS, which shows a tear in the alveolar-capillary interface from alveolar overdistension during

mechanical ventilation (i.e., volutrauma). Images from Reference 41, digitally retouched.

Atelectrauma

The decrease in lung distensibility in ARDS results in the collapse of small airways at the end of expiration, followed by reopening of the airways when the tidal volume is delivered. This cyclical opening and closing of the small airways creates shear forces that damage the airway epithelium (45). This phenomenon has been given the name *atelectrauma* (46), and it can be minimized or eliminated by applying positive end-expiratory pressure (PEEP) to hold the small airways open at the end of expiration.

Biotrauma

The physical forces involved in high-volume mechanical ventilation in ARDS can trigger the release of proinflammatory cytokines in the lungs, and these can reach the systemic circulation and produce a systemic inflammatory response (47). This occurs in the absence of structural damage in the lungs, and is known as *biotrauma*. The systemic inflammatory response from biotrauma can promote inflammatory injury in other organs, which means that *mechanical ventilation can be a source of extrapulmonary organ injury and multiorgan failure* (48)!

Lung Protective Ventilation

Lung protective ventilation employs reduced tidal volumes to protect against volutrauma and biotrauma, and uses positive end-expiratory pressure (PEEP) to prevent atelectrauma. A protocol for lung protective ventilation is shown in Table 24.5. The goal is a tidal volume of 6 mL/kg, using *predicted body weight*, which is the body weight associated with normal lung volumes. Note that another goal is an end-inspiratory “plateau” pressure ≤ 30 cm H₂O. This pressure is described in Chapter 27, and is a reflection of the peak pressure in the alveoli. A peak alveolar pressure of >30 cm H₂O creates a risk of alveolar rupture, hence the goal of lung protective ventilation is a peak alveolar pressure ≤ 30 cm H₂O.

TABLE 24.5 Protocol for Lung Protective Ventilation in ARDS

First Stage	1. Calculate patient's predicted body weight (PBW)[†]. Males: $PBW = 50 + [2.3 \times (\text{height in inches} - 60)]$ Females: $PBW = 45.5 + [2.3 \times (\text{height in inches} - 60)]$ 2. Set initial tidal volume (V_T) at 8 mL/kg PBW. 3. Add positive end-expiratory pressure (PEEP) of 5 cm H₂O. 4. Select the lowest FIO₂ that achieves an SpO₂ of 88–95%. 5. Reduce V_T by 1 mL/kg every 2 hrs until $V_T = 6$ mL/kg.
Second Stage	1. When $V_T = 6$ mL/kg, measure plateau pressure (Ppl). 2. If Ppl >30 cm H₂O decrease V_T in 1 mL/kg increments until Ppl <30 cm H₂O or $V_T = 4$ mL/kg.
Third Stage	1. Monitor blood gases for respiratory acidosis. 2. If pH = 7.15–7.30, increase respiratory rate (RR) until pH >7.30 or RR = 35 bpm. 3. If pH <7.15, increase RR to 35 bpm. If pH is still <7.15, increase V_T in 1 mL/kg increments until pH >7.15.

Optimal Goals	$V_T = 6 \text{ mL/kg}$, $P_{pl} \leq 30 \text{ cm H}_2\text{O}$, $\text{SpO}_2 = 88\text{--}95\%$, $\text{pH} = 7.30\text{--}7.45$.
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Adapted from the protocol developed by the ARDS Network, available at www.ardsnet.org.

*Predicted body weight is the weight associated with normal lung volumes.

Positive End-Expiratory Pressure

Lung protective ventilation employs a positive end-expiratory pressure (PEEP) of 5 cm H₂O, to prevent the collapse of small airways at the end of expiration. This is aimed at preventing the cyclical opening and closing of small airways that produces atelectrauma. (PEEP is described further in [Chapter 27](#)).

Permissive Hypercapnia

Low volume ventilation impairs CO₂ removal, and promotes hypercapnia. This is allowed as long as there is no evidence of harm (e.g., CO₂ narcosis). The limits of tolerance to hypercapnia and respiratory acidosis are unclear, but data from clinical trials of “permissive hypercapnia” show that arterial PCO₂ levels of 60–70 mm Hg and arterial pH levels of 7.2–7.25 are safe for most patients (49). The target pH is 7.30–7.45 in the protocol for lung protective ventilation.

Survival Benefit

Lung protective ventilation is the only ventilatory strategy that has shown a survival benefit in ARDS (49,50); as a result, it is strongly recommended as the preferred method of mechanical ventilation in ARDS (2,3). However, the benefit of lung protective ventilation is not a consistent finding (51), and it has not gained widespread acceptance. In one survey, fewer than two-thirds of patients with ARDS received tidal volumes $\leq 8 \text{ mL/kg}$ predicted body weight (1).

Airway Pressure Release Ventilation

A different approach to mechanical ventilation in ARDS is to open collapsed alveoli with a mode of ventilation known as “airway pressure release ventilation (APRV), which uses prolonged periods of spontaneous breathing at high intrathoracic pressures, interspersed with brief periods of pressure release. This is shown in [Figure 24.7](#). The prolonged periods of breathing at high pressures are designed to open collapsed alveoli, which will increase arterial oxygenation and reduce lung stiffness, while the pressure release facilitates CO₂ elimination (52). The high pressure is usually set at the last plateau pressure (not to exceed 30 cm H₂O to prevent alveolar overdistension), while the low pressure is set at 5 cm H₂O.

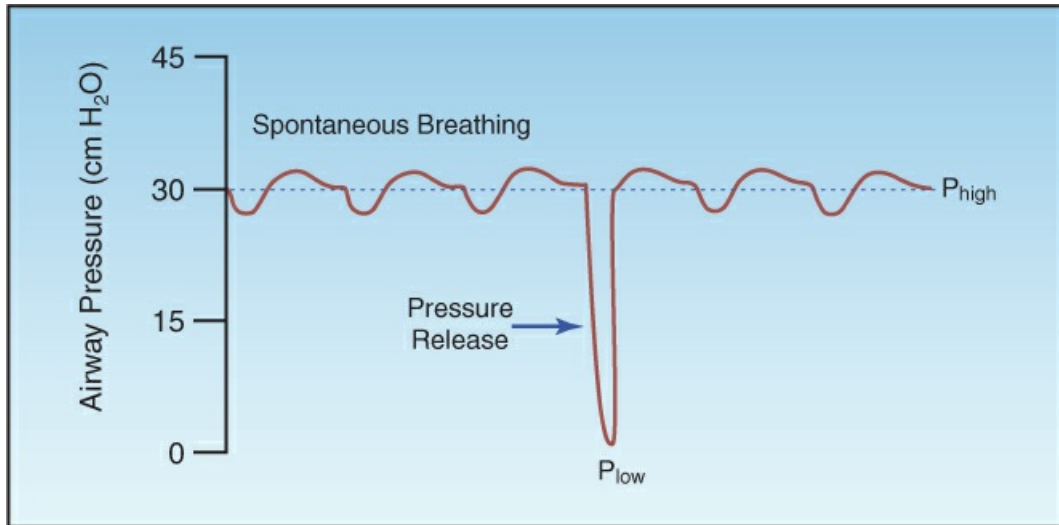


FIGURE 24.7 The pattern of ventilation in “airway pressure release ventilation” (APRV). See text for explanation.

One study comparing APRV with lung protective ventilation showed that APRV reduced the time on mechanical ventilation, but without a survival benefit (53). There are also anecdotal claims of success with APRV (the author included), but more studies are awaited.

Therapeutic Misdirection?

Mechanical ventilation is not a treatment for ARDS—in fact, the benefit of lung protective ventilation is to *do less damage* with mechanical ventilation. Moreover, the focus on the lungs in treating ARDS neglects evidence showing that *the principal cause of death in ARDS is multiorgan failure, not respiratory failure* (with the possible exception of COVID-19) (54,55). As many as 70% of deaths in ARDS are the result of multiorgan failure (54), and the mortality rate is directly related to the number of failed organs. This is demonstrated in Figure 24.8, which indicates that the mortality in ARDS is a function of a progressive systemic condition, and it implies that the therapeutic focus on the lungs in ARDS is a prescription for failure.

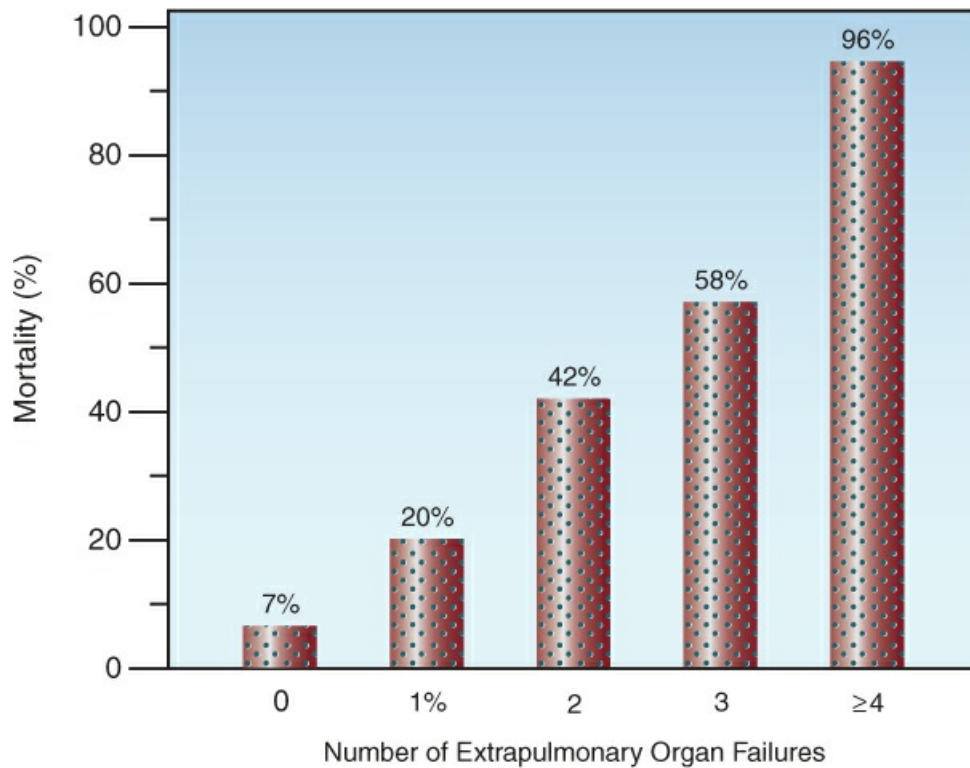


FIGURE 24.8 The relationship between extrapulmonary organ failure and mortality in ARDS. Data from Reference 54 (supplementary material).

Extracorporeal Life Support

Extracorporeal membrane oxygenation (ECMO) is a consideration for the 10–15% of ARDS patients with hypoxemia that is refractory to mechanical ventilation and prone positioning (56). The most recent indications include a $\text{PaO}_2/\text{FiO}_2$ ratio <80 mm Hg for at least 6 hours, or <50 mm Hg for at least 3 hours (57). The extracorporeal support strategy for ARDS includes venovenous ECMO with lung protective mechanical ventilation, and extracorporeal CO_2 removal, if needed. The technical aspects of extracorporeal support are beyond the scope of this text, but are available in the bibliography (55,58).

ECMO was first used for a patient with ARDS in 1971, and since that time, numerous studies have shown negative or inconsistent results, with some success in patients with viral pneumonias (e.g., influenza A(H1N1), and COVID-19) (57). Since ECMO is a temporary measure, its success is determined by the reversibility of the ARDS, which can be difficult to predict in the first few weeks of the illness.

A FINAL WORD

The Evils of Inflammation

One of the recurring themes in this book is the dominance of inflammation as a source of organ injury and lethal outcomes in critically ill patients. The destructive actions of inflammation take center stage in ARDS, and also in inflammatory shock (see Chapter 17), and it is possible that

inflammation (not hypoxia) is the principal cause of death in ICUs. (See A FINAL WORD in Chapter 17.)

As is the case with inflammatory shock, there is no specific therapy for ARDS, and management is supportive in nature. Until a remedy is available for the damaging effects of inflammation, conditions like inflammatory shock and ARDS will continue to be a major source of morbidity and mortality.

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RESPIRATORY MANAGEMENT

All who drink of this remedy will recover . . . except those in whom it does not help, who will die. Therefore, it is obvious that it fails only in incurable diseases.

Claudius Galenus
(Galen)

Oxygen Inhalation

I brought fire into your midst, and it consumed you, and reduced you to ashes.

Ezekiel (28:17)

There is a fondness for oxygen that is unmatched by any other treatment modality, as is clearly evident in the unbridled use of oxygen in hospitalized patients. The popularity of oxygen is apparent in emergency rooms, where inhaled oxygen is a knee-jerk response to acute illness, and in ICUs, where a patient who is NOT connected to a source of oxygen is a rare sight. Oxygen is given without evidence of tissue need, and without regard for the damaging effects of oxygen. This latter disregard is perplexing, since we protect our food from oxygen (with vacuum sealing, tightly-sealed plastic containers, and cellophane wrapping), yet we don't have the same consideration for our patients.

This chapter begins with an examination of the indications and effects of oxygen inhalation, and then describes the different delivery systems for oxygen. The final section is devoted to the dark side of oxygen: i.e., oxygen-related tissue injury.

INDICATIONS

The first clinical practice guideline for O₂ therapy was published in 1984 (1), and the indication for O₂ therapy was stated as follows:

Supplemental oxygen therapy is appropriate in acute conditions when there is laboratory documentation of an arterial PO₂ (PaO₂) <60 mm Hg or an arterial O₂ saturation (SaO₂) <90%; *tissue hypoxia is commonly assumed to be present at these laboratory values* [italics mine].

This statement is consistent with the consensus definition of hypoxemia as a PaO₂ <60 mm Hg or an SaO₂ <90%. The most recent guidelines for O₂ therapy (2) include a target SpO₂ (i.e., O₂ saturation by pulse oximetry) of 88–92% for patients with CO₂ retention, and 90–94% for other patients. The major issue with this recommendation is the assumption that tissue oxygenation is impaired when the PaO₂ falls below 60 mm Hg, or the SpO₂ falls below 90%. The following sections shed some light on this issue

Arterial O₂ Content

The threshold for O₂ inhalation is associated with a very small change in the arterial O₂ content. This is demonstrated in [Figure 25.1](#), which shows the arterial O₂ content (CaO₂) at the threshold for both O₂ inhalation and red blood cell (RBC) transfusions (i.e., two interventions aimed at augmenting tissue oxygenation). The CaO₂ was calculated using the equation shown below [see also Equation (9.9)]:

$$\text{CaO}_2 \text{ (mL/L)} = 1.34 \times \text{Hb} \times \text{SaO}_2 (\times 10) \quad (25.1)$$

where 1.34 is the O₂ carrying capacity of hemoglobin (in mL/g), Hb is the hemoglobin concentration (in g/dL), SaO₂ is arterial oxyhemoglobin saturation (expressed as a decimal rather than a percentage), and the factor of 10 is used to convert mL/dL to mL/L. Using normal values for Hb (15 g/dL) and SaO₂ (0.98) yields a CaO₂ of 197 mL/L, which is indicated on the left in [Figure 25.1](#). When the SaO₂ is reduced to 0.90 (the threshold for O₂ therapy), the CaO₂ is 181 mL/L, which is an 8% decrease in CaO₂. In comparison, using a Hb of 7 g/dL as the threshold for RBC transfusions (see [Chapter 12](#)), the corresponding CaO₂ is 92 mL/L, which is a 64% decrease from baseline.

The following statements can be derived from the comparisons in [Figure 25.1](#):

- . The threshold for O₂ inhalation corresponds to a minor (8%) change in arterial oxygenation.
- . Tissue oxygenation is not impaired until the CaO₂ falls below 92 mL/L (the threshold for RBC transfusions), so it cannot be impaired at the much higher threshold for O₂ inhalation.
- . It follows then, that the threshold for O₂ inhalation can be lowered.

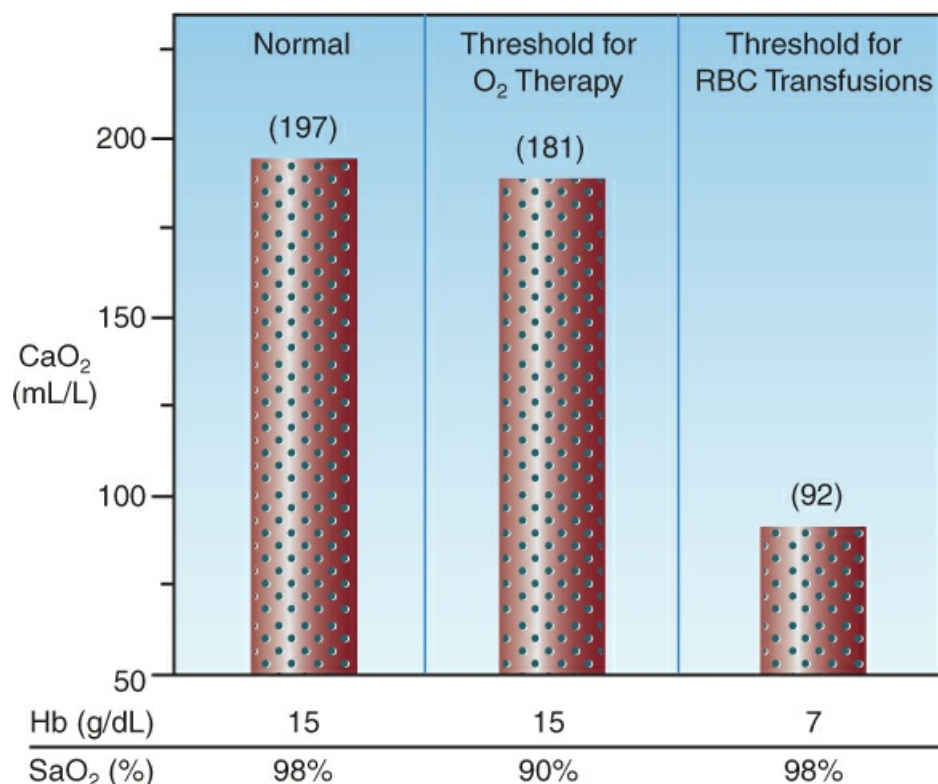


FIGURE 25.1 The arterial O₂ content (CaO₂) at the thresholds for O₂ inhalation (SaO₂ = 90%) and red blood cell transfusions (Hb = 7 g/dL).

Tolerance to Severe Hypoxemia

Since hypoxemia is corrected immediately with supplemental O₂, there is limited information on tolerance to severe hypoxemia. The available information comes from small observational studies and case reports. The data in [Table 25.1](#) is from a study of eight patients with acute exacerbation of COPD who had a PaO₂ below 40 mm Hg while breathing room air for at least one hour (3). The PaO₂, SaO₂, and plasma lactate level is listed for each patient (in ascending order according to severity of hypoxemia), and there is no evidence of impaired tissue oxygenation in any patient, as determined by the normal lactate levels (≤ 2 mmol/L). Similar observations have been reported in patients with acute respiratory distress syndrome (4), indicating that tolerance to severe hypoxemia is not an adaptation that develops over time.

TABLE 25.1 Tolerance to Severe Hypoxemia			
Patient	PaO ₂ (mm Hg)	SaO ₂ (%)	Lactate (mm/L)
1	22	35	0.9
2	30	54	0.3
3	32	59	0.9
4	33	55	1.6
5	34	65	1.6

6	35	67	2.0
7	37	75	2.0
8	39	76	1.1

From Reference 3.

Hypobaric Hypoxemia

Tolerance to severe hypoxemia was also demonstrated in a study of healthy individuals who were placed in a decompression chamber to simulate the atmospheric pressure at the Summit of Mount Everest (which is 253 mm Hg, or about one-third the pressure at sea level) (5). At this pressure, the PaO₂ was 30 mm Hg and the SaO₂ was 58% (mean values), but the plasma lactate level remained in the normal range at 1.7 mmol/L. Similar findings have been reported in climbers near the summit of Mount Everest without supplemental oxygen (6).

Reinhold Messner

One of the most notable challenges to the dogma about severe hypoxemia came from Reinhold Messner, the most celebrated mountaineer in the history of the discipline, who was the first to climb Mount Everest (elevation 29,029 feet) without the use of supplemental oxygen (a feat he achieved in 1978, along with fellow climber Peter Habeler). Prior to the historic climb, Messner was warned by experts that attempting to summit Mount Everest without supplemental O₂ would be suicidal, as he would lose consciousness and suffer permanent brain damage above an altitude of 26,000 feet (the region known as the “death zone”). However, Messner knew this was an untested claim (like many about oxygen), so he devised a simple test: i.e., using a plane with a depressurized cabin, he was flown to the summit of Mount Everest while breathing ambient air. To the disbelief of the experts, Messner experienced no ill effects as the plane climbed above 26,000 feet (the death zone) and reached its destination. He later wrote, “I had seen the flight through without an oxygen mask, and I was still able to talk, to think, to sense everything” (7).

Implications

In summary, there is considerable evidence indicating that *O₂ inhalation is not based on tissue O₂ needs*; a notion that is supported by the physiological effects of O₂ described in the next section. As a result, the use of O₂ inhalation based on current recommendations results in excessive oxygenation, and this can be detrimental by promoting oxidative tissue injury, as described in the latter part of the chapter.

PHYSIOLOGICAL EFFECTS

Metabolic Rate

The goal of O₂ inhalation is to promote aerobic metabolism. However, there is no evidence that aerobic metabolism is impaired at the threshold for O₂ inhalation, so it seems unlikely that O₂ inhalation will have any impact on aerobic metabolism. This is demonstrated in Figure 25.2,

which shows the effects of breathing 24% and 28% oxygen on the arterial PO_2 (PaO_2) and the rate of aerobic metabolism (VO_2) in patients with acute exacerbation of chronic obstructive lung disease (8). There is a significant increase in the PaO_2 with each increment in inhaled O_2 , but the rate of aerobic metabolism remains unchanged. Similar results have been reported in other clinical studies (9,10), indicating that *oxygen inhalation does not promote aerobic metabolism*.

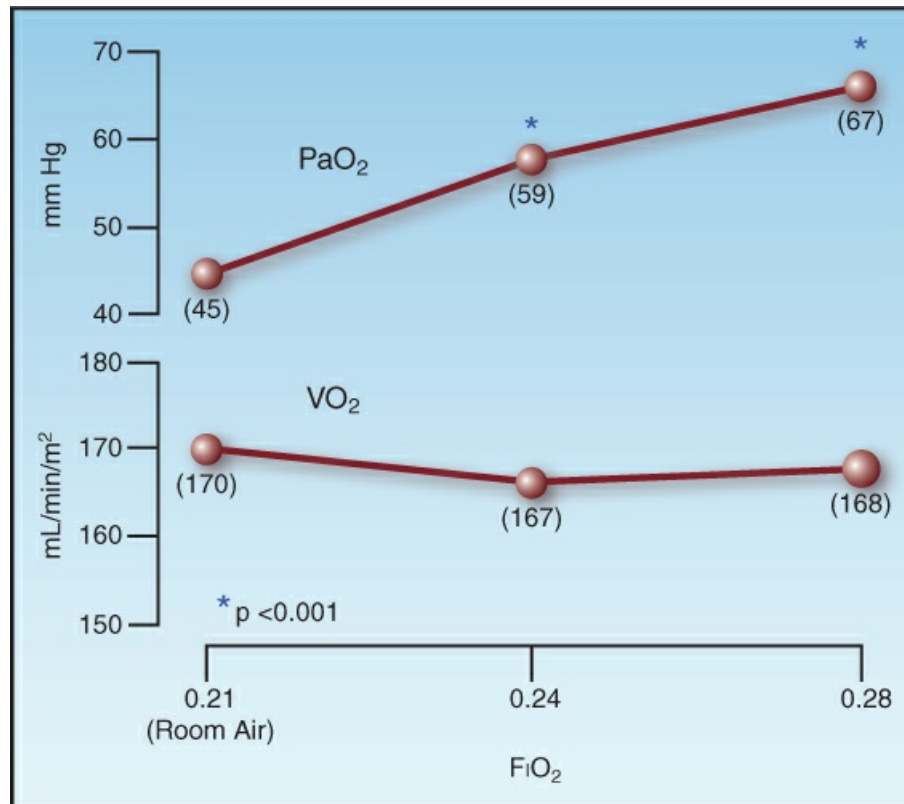


FIGURE 25.2 Results of a study in patients with severe hypoxemia showing that the increase in arterial PO_2 (PaO_2) from O_2 therapy is not accompanied by an increase in aerobic metabolism (VO_2). Numbers in parentheses are the mean values for each measurement. FiO_2 = fraction of O_2 in inhaled gas. Data from Reference 8.

Oxygen as a Vasoconstrictor

Oxygen acts as a vasoconstrictor in all major organs except the lungs (11), where it acts as a vasodilator. The principal mechanism for the vasoconstriction is loss of the vasodilating actions of nitric oxide, which is oxidized by one of the oxygen metabolites, the superoxide radical (see later) (12). This vasoconstriction has the following consequences:

- . Oxygen promotes vasoconstriction in the coronary arteries, and can decrease coronary blood flow in patients with coronary artery disease (13).
- . Oxygen-induced systemic vasoconstriction can decrease the cardiac output and promote tissue hypoperfusion, and this effect counteracts the ability of O_2 inhalation to increase systemic O_2 delivery (14).
- . In animal studies, oxygen can completely obliterate capillary networks in skeletal muscle (15).

Implications

The observation that O₂ inhalation does not promote aerobic metabolism is consistent with the notion that the clinical use of O₂ is not based on tissue O₂ needs. Furthermore, *oxygen-induced vasoconstriction can be viewed as an inherent mechanism that protects the tissues from excess (unwanted) oxygen*. In other words, the tissues have enough O₂, and they don't want anymore.

OXYGEN DELIVERY SYSTEMS

A variety of oxygen delivery systems are available for spontaneously breathing patients, and the major ones are listed in [Table 25.2](#), along with some characteristic features of each.

Flow Rate and FIO₂

Supplemental O₂ is provided as a pure gas that is delivered into the mouth or nasopharynx at a preselected flow rate, and the fractional concentration of inhaled O₂ (FIO₂) is determined by the balance between the rate of O₂ delivery and the patient's inspiratory flow rate. (There are some exceptions to this, as described later.) If the patient's inspiratory flow rate exceeds the rate of O₂ delivery, the excess flow will draw in room air, which will decrease the FIO₂. A normal inspiratory flow rate is about 15 L/min (0.25 L/sec) during quiet breathing, but it can be as high as 120 L/min in patients with acute respiratory failure (16). Therefore, low-flow O₂ delivery systems (which have delivery rates below 15 L/min) are usually inadequate for patients with acute respiratory failure, especially those with respiratory distress or labored breathing.

Low-Flow Systems

Low-flow systems deliver O₂ at rates of ≤15 L/min, and are most suitable for patients with mild hypoxemia and no respiratory distress.

Low-Flow Nasal O₂

Low flow O₂ is typically delivered via nasal prongs at flow rates of 1–6 L/min (although rates up to 15 L/min are possible with large-bore nasal cannulas). The O₂ is usually humidified when the flow rate exceeds 4 L/min. The FIO₂ during quiet breathing is about 24% (at 1 L/min) up to 40% (at 6 L/min).

The major advantages of low-flow nasal O₂ are simplicity of use and patient acceptance, including the ability for patients to eat and converse. The major disadvantage is the inability to achieve high concentrations of inhaled O₂.

Standard Face Masks

Standard face masks deliver oxygen at flow rates between 5 and 10 L/min (a minimum flow rate of 5 L/min is needed to clear exhaled gas from the mask). Exhalation ports on the side of the face mask also allow room air to be inhaled. This system can achieve a maximum FIO₂ of about 60% during quiet breathing. Although face masks can deliver a slightly higher FIO₂ than nasal prongs, they are more confining, and they do not permit oral feeding.

TABLE 25.2

Oxygen Delivery Systems

System or Device	Flow Rates	FIO ₂ Range	Comments
Low-Flow Nasal O ₂	1–6 L/min	24–40%	Used for mild hypoxemia with no respiratory distress.
Standard Face Mask	5–10 L/min	35–50%	Same as above.
Non-Rebreather Mask	≥10 L/min	60–80%	Used for high O ₂ requirements, but no longer favored.
Air-Entrainment Mask	2–15 L/min	24–50%	Designed for controlled O ₂ delivery.
OxyMask™	1–≥15 L/min	24–90%	Can deliver high O ₂ concentrations at relatively low flow rates.
High-Flow Nasal O ₂	1–60 L/min	24–100%	Most effective system for severe hypoxemia and respiratory distress.

High-Flow Systems

High-flow systems include reservoir masks, air-entrainment devices, specialized “diffuser” masks (OxyMask™), and high-flow nasal O₂. These systems are intended for patients with high O₂ requirements and high ventilatory demands.

Reservoir Masks

Reservoir masks are standard face masks that are attached to a reservoir bag that is filled with oxygen from a constant-flow oxygen source. If the reservoir bag is kept inflated, the patient will draw primarily from the gas in the bag, which allows for higher concentrations of inhaled oxygen.

An example of a reservoir mask is shown in [Figure 25.3](#). This is a *non-rebreather mask*, which is designed to prevent rebreathing of exhaled CO₂. The reservoir bag is filled with oxygen, and a one way valve between the bag and the mask allows inhalation of O₂ from the bag, but prevents exhaled CO₂ from entering the bag. One-way flaps on the mask allow exhaled gas to escape, while preventing inhalation of ambient air. Non-rebreather masks can theoretically achieve an FIO₂ of 100%, but the maximum FIO₂ is closer to 80%, because of leaks around the mask and valve malfunction. (*Note: There is also a *partial rebreather mask*, which allows some of the exhaled CO₂ to enter the reservoir bag, but these devices are rarely used.*)

Non-rebreather masks were once a popular method for providing high concentrations of O₂, but they have fallen from favor because of problems with CO₂ rebreathing from faulty valves, and troublesome hypoxemia from inadequately filled reservoir bags.

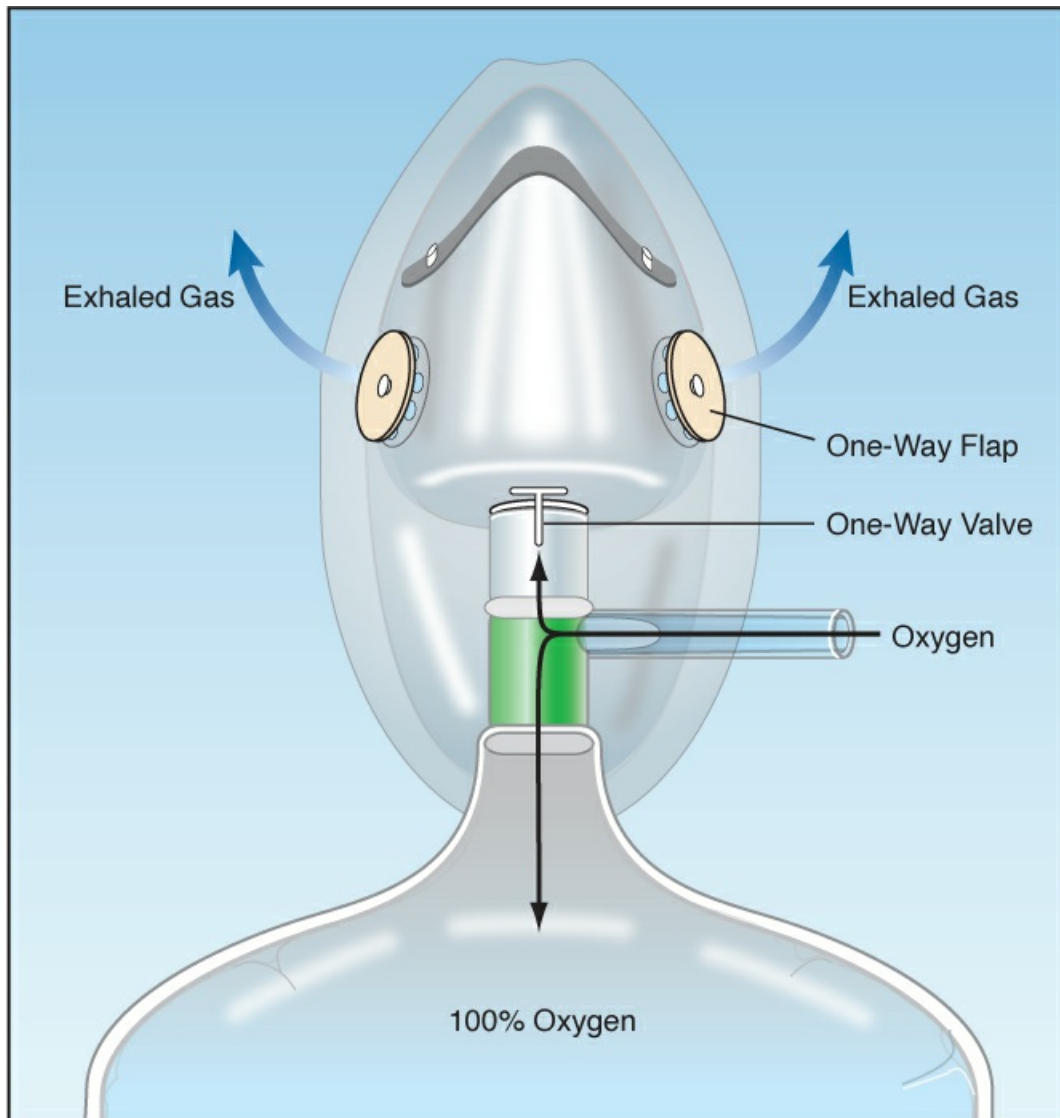


FIGURE 25.3 Non-rebreather mask. Oxygen is inhaled from the reservoir bag, and a one way valve prevents exhaled gas from entering the bag. Flaps on the mask allow exhaled gas to escape, while preventing inhalation of ambient air.

Air Entrainment Masks

Air entrainment masks are designed to deliver a constant FiO_2 . The operation of an air-entrainment device is shown in Figure 25.4 (17). The end of the oxygen inlet port is narrowed to create a high velocity stream of gas (analogous to the nozzle on a garden hose). This produces a shear force known as *viscous drag* that pulls room air into the device through air-entrainment ports. The greater the flow of O_2 into the mask, the greater the volume of air that is entrained, and this keeps the FiO_2 constant. The FiO_2 is varied by varying the size of the air entrainment ports. Although air-entrainment masks use O_2 flow rates of ≤ 15 L/min, they are considered high-flow systems because the total flow (i.e., O_2 flow plus room air flow) is often ≥ 60 L/min. The FiO_2 range of these devices is 24 to 50%.

Air-entrainment masks have been popular for patients with chronic CO_2 retention, where an

inadvertent increase in FiO_2 can lead to further increases in arterial PCO_2 . However, their popularity has waned, because the continuous monitoring of arterial O_2 saturation with pulse oximetry is a deterrent to excessive oxygenation in patients with CO_2 retention.

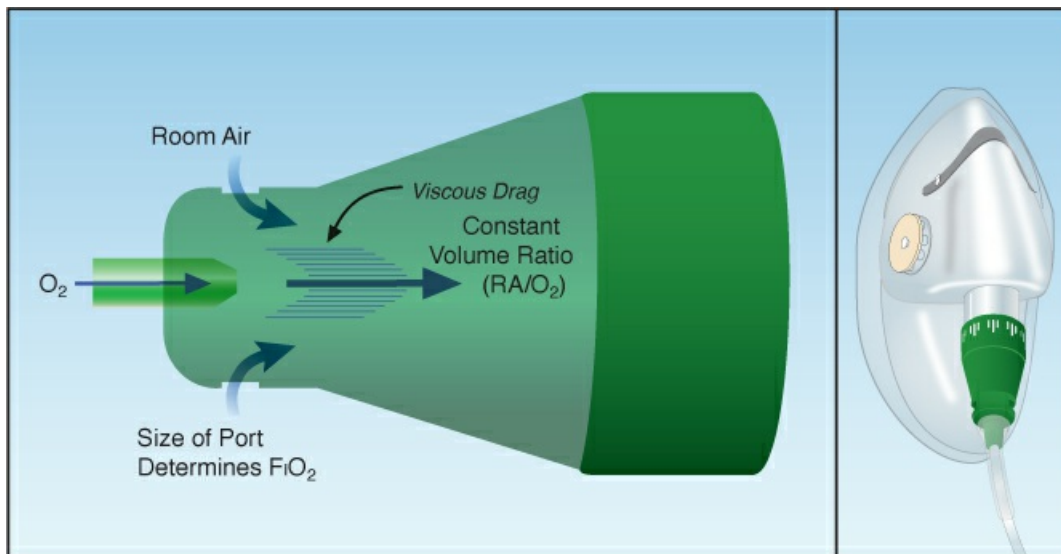


FIGURE 25.4 An air-entrainment device. A narrowing at the oxygen inlet creates a high-velocity stream of gas that creates viscous drag, which pulls in room air (RA). This “jet mixing” keeps the concentration of inhaled oxygen constant, regardless of changes in the flow rate of oxygen.

The OxyMask™

The OxyMask™ (Southmedic, Inc., Barrie, Ontario, Canada) is equipped with a proprietary device called a “diffuser” that delivers a concentrated stream of O_2 directly over the nose and mouth. This design allows for high concentrations of inhaled O_2 at relatively low flow rates (see Table 25.2). The mask also has large openings to facilitate the escape of exhaled CO_2 .

The OxyMask has outperformed the non-rebreather mask (less CO_2 rebreathing) (18) and the air-entrainment mask (greater O_2 concentration at lower flows) (19). It is a superior mask for delivering high concentrations of O_2 , at a relatively low cost (because of the low flow rate for O_2). However this mask otherwise offers no advantages over the high-flow nasal O_2 system described next.

High-Flow Nasal O_2

High-flow nasal O_2 (HFNO) is the greatest advance in O_2 delivery systems in recent memory. The basic design of the HFNO circuit is illustrated in Figure 25.5. This circuit provides heated and humidified gas for inhalation, and has an oxygen-air blender that allows titration of the FiO_2 and flow rate independently. Flow rates of 40–60 L/min are achievable (depending on the manufacturer), and the nasal prongs have a tapered end to increase the velocity of flow (like the nozzle on a garden hose).

Physiological Benefits

HFNO has a number of beneficial effects, and these are listed in Table 25.3. The high flow rates,

combined with the increased velocity of flow from the design of the nasal prongs, results in a washout of the dead space in the nasopharynx (20,21). (Note: About 30% of a tidal volume is drawn from exhaled gas in the anatomical dead space.) This prevents rebreathing of exhaled CO₂, and also provides a “sink” that maintains the desired FIO₂ if a patient overbreathes the flow rate for O₂ delivery. Surprisingly, HFNO also increases the end-expiratory lung volume, which, in turn, increases lung compliance (distensibility) and reduces the work of breathing (20,21).

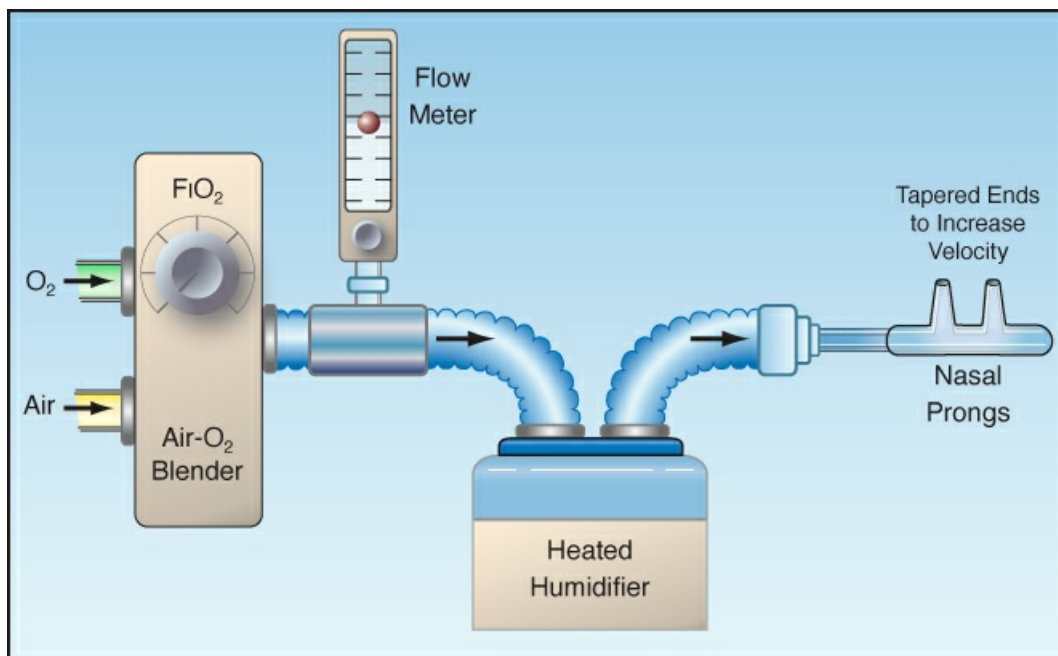


FIGURE 25.5 A schematic illustration of the circuit for high-flow nasal O₂. See text for explanation.

The ability of HFNO to increase the end-expiratory lung volume has been attributed to positive end-expiratory pressure (PEEP). However, HFNO can generate low-level PEEP (2–3 cm H₂O), but only with the mouth closed (22,23), which is a rare phenomenon in patients with acute respiratory failure.

TABLE 25.3 The Benefits of High-Flow Nasal Oxygen	
Physiological	Clinical
• Reduces dead space • Increases end-expiratory lung volume • Increases lung distensibility • Reduces respiratory rate and minute ventilation • Reduces work of breathing	• Most effective O₂ delivery system for severe hypoxemia • Promotes comfortable breathing • Can reduce the risk of “patient self-induced lung injury” • Allows oral food intake • Can prevent intubation

Clinical Benefits

Other than the ability to correct severe hypoxemia, the principal benefit of HFNO is the tendency to alleviate respiratory distress and promote comfortable breathing. This not only promotes

patient comfort, it also reduces the risk of “patient self-induced injury”, a form of lung injury produced by labored breathing, which is described in [Chapter 24](#) (see [Figure 24.4](#)). This may explain why HFNO has reduced the rate of intubations in some studies ([24](#)), although this is not a consistent finding ([25](#)).

HFNO has traditionally been used when hypoxemia is refractory to conventional methods of O₂ inhalation, but the numerous benefits of HFNO has prompted a recommendation that HFNO should *replace* conventional O₂ therapy for patients with acute hypoxemic respiratory failure ([26](#)).

OXYGEN AS A SOURCE OF INJURY

The introduction to this chapter included a statement about protecting our food from the oxygen in atmospheric air (e.g., with vacuum sealing), and the reason is that *oxygen decomposes organic matter* by disrupting organic molecules (including carbohydrates, proteins, and lipids). The chemical derivatives of oxygen, which are called *reactive oxygen species* (ROS) are even more damaging than the parent molecule, and are capable of inflicting lethal cell injury ([27](#)). In fact, contrary to the popular perception that oxygen protects cells from injury in critically ill patients, the accumulating evidence indicates that *oxygen (via the production of ROS) is an important source of cell injury in critically ill patients* ([27–29](#)). (The role of ROS in inflammation is described in [Chapter 17](#).) The following is a brief description of the injurious nature of oxygen.

Oxygen Metabolism

The metabolism of O₂ takes place at the end of the electron transport chain in mitochondria (within the cytochrome oxidase complex), where the electrons that accumulate in the process of ATP production are cleared by adding them to O₂ to produce water. (The addition of electrons to an atom or molecule is known as chemical reduction.) The O₂ molecule can accept four electrons, but only one electron can be added per reduction reaction (the reason for this is beyond the scope of this chapter), which results in the reaction sequence shown in [Figure 25.6](#).

The intermediates in the reduction of O₂ to water include the superoxide radical, hydrogen peroxide, and the hydroxyl radical. All are reactive oxygen species (ROS) that are capable of disrupting and damaging vital cell components (e.g., membrane lipids, cytoplasmic proteins, and DNA) by the chemical process of *oxidation* (which removes electrons from a substrate). Thus, ROS are sometimes referred to as oxidizing agents, or *oxidants*. Some relevant features of the ROS generated by O₂ metabolism are summarized below ([27](#)).

- . The superoxide and hydroxyl radicals are *free radicals* (i.e., they have an unpaired electron in their outer orbitals), and are highly reactive.
- . Hydrogen peroxide is not a free radical, but it is a powerful oxidizing agent that readily moves across cell membranes, and can be widely distributed throughout the body.
- . The hydroxyl radical is the most reactive molecule known in biochemistry, and enters into a reaction within three molecular diameters of its point of origin. It is the most destructive of all the ROS, and is a major source of oxidant-induced cell injury.

- . Free iron in its reduced form (Fe^{++}) catalyzes the formation of the hydroxyl radical, and thus free iron can act as a pro-oxidant (see later).

Normally, at least 95% of oxygen is completely reduced to water, and only a very small fraction (about 3%) of O_2 metabolism generates damaging ROS (27). However, there is *a more important, source of ROS: i.e., the inflammatory response* (see next).

Neutrophil Activation

As described in Chapter 17 (see Figure 17.1), the activation of neutrophils involves a precipitous rise (up to 20-fold) in O_2 consumption (30), which is known as the *respiratory burst*, and is designed to generate ROS. Neutrophils also have a myeloperoxidase enzyme that generates *hypochlorite*, which is the active ingredient in household bleach.

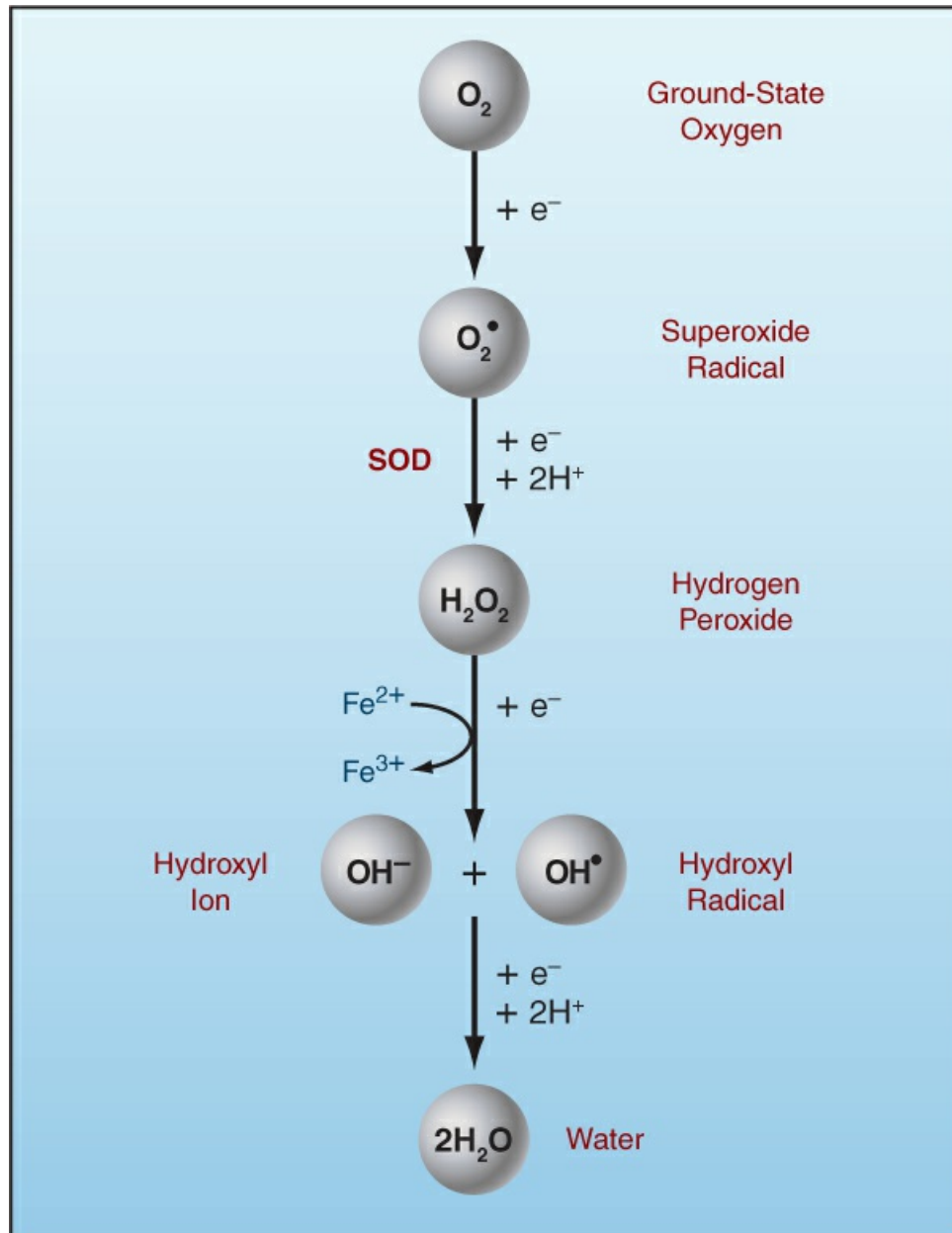


FIGURE 25.6 The reaction sequence for the chemical reduction of oxygen to water, which takes place at the end of the electron transport chain in mitochondria. Free radicals are indicated by a superscripted dot. See text for explanation.

Inflammatory Injury

There is abundant evidence that ROS have a prominent role in the tissue injury attributed to inflammation (see [Chapter 17](#)) (31). The damaging effects of ROS are accelerated by the *tendency for free radicals to create chain reactions* (32). When a free radical reacts with a non-radical, the non-radical loses an electron and is transformed into a free radical, which can then remove an electron from another nonradical to produce another free radical, and so on. This creates a self-sustaining reaction or *chain reaction*, and these reactions are troublesome because they continue after the inciting event is eliminated. (A fire is a familiar example of a chain

reaction involving free radicals.) Chain reactions would explain the progression of inflammatory multiorgan damage in severe sepsis and septic shock after the infection has been eradicated.

Antioxidant Protection

Oxidative (oxygen-related) injury is kept in check by a vast array of endogenous *antioxidants* (i.e., atoms or molecules that prevent or block the actions of oxidizing agents). The following is a brief description of the major antioxidants, and Figure 25.7 shows their mechanisms of action.

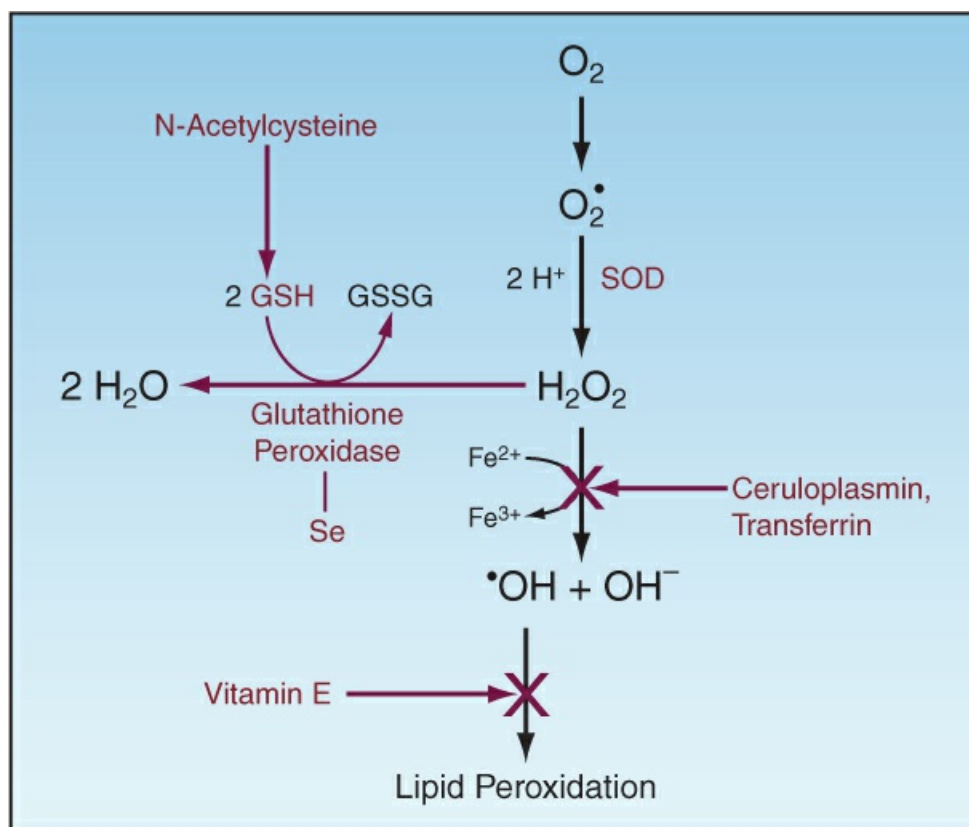


FIGURE 25.7 The actions of some endogenous and exogenous antioxidants (shown in red). SOD = superoxide dismutase, GSH = reduced glutathione, GSSG = oxidized glutathione, Se = selenium. See text for explanation.

Superoxide Dismutase

Superoxide dismutase (SOD) is an enzyme that facilitates the conversion of superoxide radicals to hydrogen peroxide. SOD is considered an antioxidant because it clears the superoxide radical, but it also increases the production of hydrogen peroxide, and thus it can also act as a pro-oxidant (33).

The role of SOD in inflammatory injury is suggested by animal studies of pain perception, which have shown that treatment with an SOD mimetic curtails the pain response to inflammation (34). This implicates superoxide radicals as a source of inflammatory pain; a notion that is supported by evidence that SOD is inactivated by products of inflammation (35). These findings have important implications for the development of nonopioid analgesics for painful inflammation.

Glutathione

Glutathione is a sulfur-containing tripeptide that is considered *the major intracellular antioxidant in the human body* (36,37). In its reduced form, glutathione (GSH in Figure 25.7) donates electrons to reduce hydrogen peroxide to water in a reaction that is catalyzed by a selenium-dependent enzyme, glutathione peroxidase:



Glutathione is present in high concentrations in most mammalian cells, and is synthesized *de novo* within cells. It can be exported extracellularly, but plasma levels are three orders of magnitude lower than intracellular levels (38). Glutathione levels in lung lavage fluid are 140-fold higher than plasma (39), suggesting that glutathione plays an important role in protecting the lung from oxidant injury. There is evidence that *glutathione is depleted in critically ill patients* (40).

N-ACETYLCYSTEINE: Glutathione does not move readily into cells, and exogenous glutathione administration has little effect on intracellular levels (41). However, the popular mucolytic agent *N-acetylcysteine* is a glutathione analogue that can cross cell membranes and serve as a glutathione surrogate (as proven by the beneficial effects of N-acetylcysteine in acetaminophen toxicity, which is attributed to glutathione deficiency - see Chapter 52). N-acetylcysteine has a promising future as an antioxidant in critically ill patients (42).

SELENIUM: Selenium is an essential trace element that serves as a cofactor for the glutathione peroxidase enzyme in humans. The recommended dietary allowance (RDA) for selenium is 55 µg daily in adult men and women (43). The absence of dietary selenium produces measurable decreases in glutathione peroxidase activity after just one week (44). Selenium levels in blood are typically low in critically ill patients (45), and high-dose selenium replacement (1000 µg IV daily) has been associated with improved survival in patients with severe sepsis and septic shock (45).

Selenium status can be monitored using whole blood selenium (normal range = 0.96–1.78 µmol/L) or serum selenium (normal range = 0.72–1.33 µmol/L). If needed, selenium can be given intravenously as sodium selenite. The highest dose that is considered safe is 200 µg daily.

Vitamin E

Vitamin E (alpha-tocopherol) is a lipid-soluble vitamin that is found in the interior of most cell membranes, where it serves as a “chain breaking” antioxidant to halt the progression of lipid peroxidation (i.e., the oxidation of polyunsaturated fatty acids, also known as *rancidity* when it occurs in food products), which proceeds as a chain reaction. To accomplish this, vitamin E donates an electron to a free radical intermediate in lipid peroxidation, and in so doing, vitamin E becomes a free radical, but an innocuous one. This reaction halts the chain reaction of lipid peroxidation. (*Note:* Lipid peroxidation is initiated by the hydroxyl radical, and Figure 25.7 shows Vitamin E blocking this reaction, to indicate that Vitamin E blocks lipid peroxidation.)

Vitamin E depletion has been reported in patients with acute respiratory distress syndrome (46), and there is evidence that high-dose vitamin E has beneficial effects in trauma victims (see later). The normal concentration of vitamin E in plasma is 1 mg/dL, and a level below 0.5 mg/dL

is evidence of deficiency (47).

Vitamin C

Vitamin C (ascorbic acid) is a water-soluble vitamin that is best known for its essential role in collagen formation. However, it is also a reducing agent that can donate electrons to free radicals to erase their adverse effects. As a result, ascorbate can act as a “scavenger” for oxygen-derived free radicals: i.e., superoxide radicals and hydroxyl radicals (48). Ascorbate also donates electrons to regenerate Vitamin E from the inactive Vitamin E radical.

Ceruloplasmin and Transferrin

Ceruloplasmin and transferrin account for most of the antioxidant activity in plasma (49). The antioxidant activity of both proteins is related to their actions in limiting free iron in the reduced form (Fe^{2+}), which will limit the production of hydroxyl radicals. Ceruloplasmin oxidizes iron from the Fe^{2+} to the Fe^{3+} state, and transferrin binds iron in the oxidized or Fe^{3+} state. The role of free iron as a pro-oxidant (50) may explain why most of the iron in the body is bound to proteins or sequestered (e.g., in the bone marrow).

Oxidative Stress

The risk of oxidative tissue injury is determined by the balance between oxidant and antioxidant activities. When oxidant activity exceeds the neutralizing capacity of the antioxidants, the excess or unopposed oxidant activity can promote tissue injury. This condition of *unopposed biological oxidation is known as oxidative stress*, and it is the principal source of oxidative tissue injury. The most familiar example of oxidative stress is severe or persistent inflammation, especially in critically ill patients, which combines an increase in production of oxidants (ROS) with likely depletion of antioxidants.

Monitoring oxidative stress has obvious benefits for the care of critically ill patients (or any patient with inflammatory tissue injury). One promising method involves the measurement of lipid peroxidation products in exhaled gas (51), but this methodology is not currently available in clinical practice.

PULMONARY OXYGEN TOXICITY

Pulmonary oxygen toxicity is an inflammatory lung injury (similar to the acute respiratory distress syndrome described in Chapter 24) that occurs in response to the prolonged inhalation of high concentrations of oxygen. The following are some relevant issues regarding this form of oxidative injury.

Species Differences

The tendency to develop pulmonary oxygen toxicity varies in different species. For example, laboratory rats will die of respiratory failure after 5 to 7 days of breathing 100% O_2 , while cold-blooded species like sea turtles can breathe pure O_2 for several weeks and even months without harm (52). Non-human primates, which are closest to humans, show signs of pulmonary O_2 toxicity after about one week of breathing pure oxygen (53).

Human Studies

In healthy volunteers, inhalation of 100% O₂ for 6 to 12 hours results in a tracheobronchitis and a decrease in vital capacity attributed to absorption atelectasis (54). Prolonged exposure to 100% O₂ has been reported in only 6 humans: 5 with irreversible coma who received 100% O₂ for 3 to 4 days (55), and one healthy volunteer who inhaled pure O₂ for 4.5 days (56). In all these cases, the subjects developed a pulmonary condition that was consistent with inflammatory lung injury.

What FIO₂ is Toxic?

Based on the observation of a decreased vital capacity when the FIO₂ exceeds 60% (54), the threshold FIO₂ for pulmonary O₂ toxicity was set at 60%. However, adopting a single threshold FIO₂ for all patients neglects the contribution of endogenous antioxidants to the risk of oxygen toxicity. For example, animal studies have shown that pulmonary O₂ toxicity is exacerbated by vitamin E deficiency (57) and glutathione deficiency (58). *Since antioxidant depletion seems to be common in ICU patients (40,45,46), it is reasonable to assume that the toxic level of FIO₂ is much lower than suspected in critically ill patients.* Therefore, the best practice is keeping the FIO₂ at the lowest tolerable level.

Promoting Antioxidant Protection

Another approach to minimizing the risk of pulmonary O₂ toxicity (and any source of oxidative cell injury) is to monitor and correct antioxidant levels. This approach is supported by evidence of antioxidant depletion in critically ill patients (40,45,46). Evidence of benefit from such an approach is provided by a study in trauma patients (59), which showed that a high-dose antioxidant cocktail of vitamin C (1000 mg every 8 hrs), vitamin E (1000 units every 8 hrs) and selenium (200 µg daily) for 7 days was associated with a significant decline in cases of respiratory failure and ventilator dependence. Promoting antioxidant protection for your patients is similar to the use of tightly sealed plastic containers to protect your food.

A FINAL WORD

Why is Oxygen a Vasoconstrictor?

For those who like to consider teleology in biological design, the actions of oxygen to promote systemic vasoconstriction merits some attention.

The ability of inhaled O₂ to augment tissue oxygenation is determined by its influence on both the O₂ content in arterial blood (CaO₂), and the cardiac output (CO), as defined by the equation for arterial O₂ delivery: i.e.,

$$\text{O}_2 \text{ Delivery} = \text{CaO}_2 \times \text{CO} \quad (25.3)$$

Inhaled O₂ will increase the CaO₂, but the O₂-induced systemic vasoconstriction leads to a decrease in CO, and the net effect is that O₂ delivery remains unchanged (60). This would explain why O₂ inhalation has no effect on aerobic metabolism, as shown in Figure 25.2.

The vasoconstrictor response to O_2 can then be viewed as a defense mechanism that is designed to protect the tissues from excess oxygen (i.e., more than is needed to maintain aerobic metabolism). This will limit the risk of oxidative tissue injury, and in this sense, *the vasoconstrictor response to O_2 can be viewed as an antioxidant defense mechanism.*

The human body seems designed to protect the vital organs from oxygen, and *our liberal use of oxygen is the contrary to human design.*

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Noninvasive Ventilation

We...repeatedly enlarge our instrumentalities without improving our purpose.

Will Durant ([a](#))

Noninvasive positive pressure breathing was introduced in the 1940s (about a decade before conventional mechanical ventilation), but it was delivered through a mouthpiece, and could only be used for brief periods of time (10–15 minutes) ([1](#)). For the next 40 years, this method of “intermittent positive pressure breathing” (IPPB) was used for delivering inhaled bronchodilators, and for the prevention of postoperative atelectasis. The use of “continuous positive airway pressure” (CPAP) was introduced in the early 1980s, and rapidly gained traction as a treatment for obstructive sleep apnea ([2](#)). This led to improvements in mask design, and by the 1990s, reports began to appear about the use of positive pressure breathing through face masks to manage patients with acute respiratory failure ([3](#)). Since that time, the popularity of *noninvasive ventilation* has skyrocketed, but the benefits have been variable (as described in this chapter).

This chapter describes the fundamentals of using noninvasive ventilation, including what to use, what to monitor, and what to expect in different types of acute respiratory failure ([4](#)). The last section describes the effects of positive pressure breathing on cardiac performance.

METHODS OF NONINVASIVE VENTILATION

The following is a description of the principal methods of pressure-assisted breathing that do not require endotracheal intubation.

Continuous Positive Airway Pressure

During normal breathing, the descent of the diaphragm creates a negative intrathoracic pressure that draws air into the lungs, and exhalation proceeds passively until the intrathoracic pressure returns to atmospheric (zero reference) pressure). When breathing with “continuous positive airway pressure” (CPAP), the pressure at the end of expiration is positive relative to atmospheric pressure, and the positive pressure is maintained through the respiratory cycle. This is

demonstrated in [Figure 26.1](#), which shows a spontaneous breathing pattern with a positive end-expiratory pressure of 5 cm H₂O.

The positive pressure in CPAP is created by a constant flow of air or oxygen that is adjusted to achieve the desired pressure level (usually 5–10 cm H₂O). This pressure helps to prevent collapse of the small airways and alveoli at the end of expiration, and increases the end-expiratory lung volume (i.e., the functional residual capacity), but it does not augment the tidal volume.

Clinical Uses

The principal use of CPAP is in patients with obstructive sleep apnea, where the positive pressure prevents the inspiratory collapse of the pharynx that causes the obstruction to airflow (5). CPAP has also been successful in managing patients with acute cardiogenic pulmonary edema (6), for reasons described later in the chapter. Overall, CPAP provides only limited pressure support, and it is rarely used to manage patients with acute respiratory failure.

Bi-Level Positive Airway Pressure

Noninvasive ventilation is typically a patient-triggered, pressure-targeted mode of ventilation that provides pressure-augmented tidal volumes along with positive end-expiratory pressure (PEEP) (7). This is illustrated in [Figure 26.1](#). The patient triggers each positive-pressure lung inflation (indicated by the negative pressure swings), and the inspiratory pressure gradually rises until it reaches a pre-selected pressure. Exhalation then proceeds until the pressure reaches a pre-selected positive end-expiratory pressure (PEEP). Peak inspiratory pressures are typically 10–20 cm H₂O, and PEEP levels are usually 5–10 cm H₂O. Higher pressure are generally not advised because they are poorly tolerated by patients, and they promote leaks around the face mask.

(*Note:* The mode of ventilation just described is typically called *bi-level positive airway pressure*, or simply *BiPAP*, however this is a proprietary name (registered as a trademark name by Philips Respironics), and is used sparingly in this chapter.

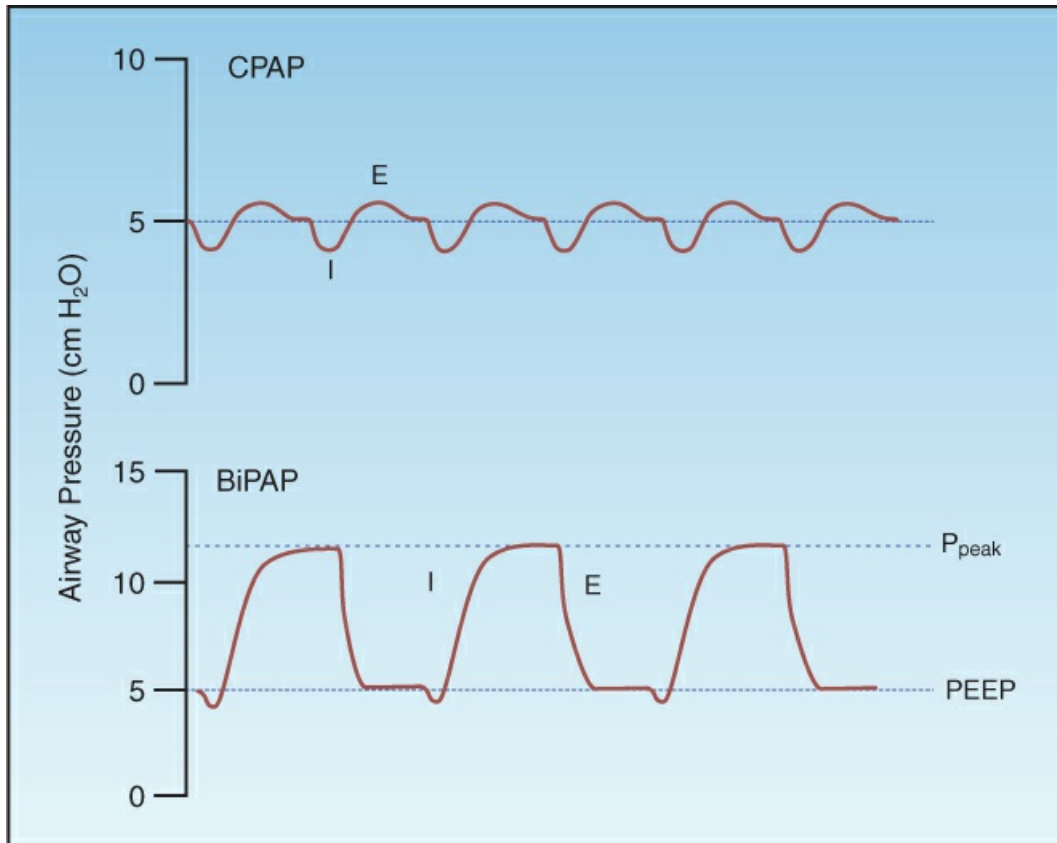


FIGURE 26.1 Airway pressure waveforms for continuous positive airway pressure (CPAP) and bi-level positive airway pressure (BiPAP). See text for explanation.

Pressure Support Ventilation

Pressure support ventilation (PSV) is an interactive form of pressure-targeted ventilation that adjusts the ventilator breath to the ventilatory demands of the patient (7,8). This is explained using the pressure waveform in Figure 26.2. The following statements correspond to the numbers located at different points on the waveform.

- . PSV is a patient-triggered mode of ventilation, and the initial negative pressure deflection represents the spontaneous inspiratory effort that triggers the ventilator breath.
- . Once the ventilator breath is triggered, high flow rates are used to produce a rapid rise in airway pressure, in order to reach the desired inspiratory pressure as soon as possible.
- . When the desired pressure level is achieved, it is kept constant by servo-adjusting the ventilator flow rate to the inspiratory flow demands of the patient.
- . The lung inflation is terminated when the patient's inspiratory flow rate drops to 25% of the peak flow.

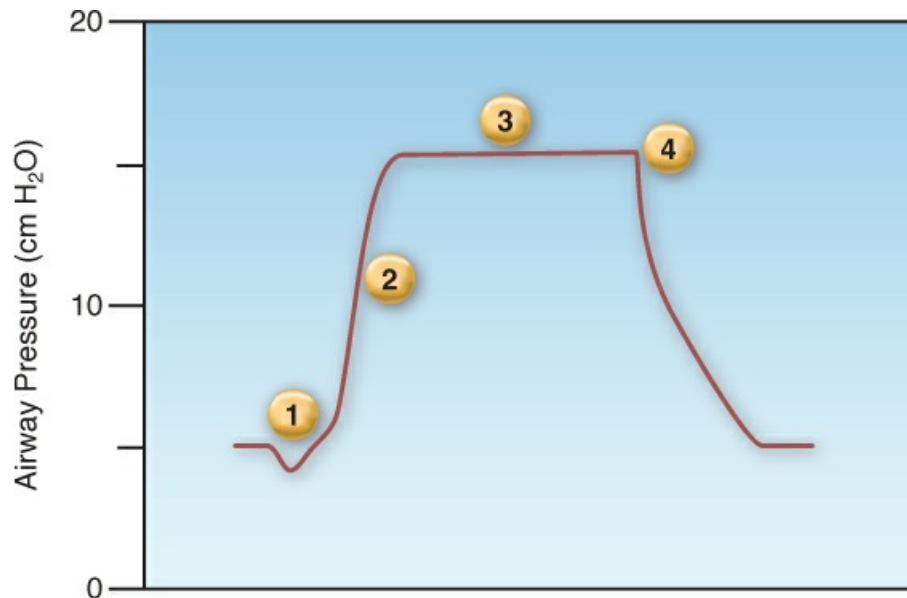


FIGURE 26.2 The pressure waveform in pressure support ventilation (PSV). The numbers at different points on the waveform are explained in the text.

This interaction allows the patient to determine the inflation time and tidal volume, which promotes patient-ventilator synchrony, and reduces the work of breathing (8). PSV is typically reserved for cases of patient-ventilator asynchrony, and is also popular for spontaneous breathing trials used in the process of weaning patients from mechanical ventilation (see [Chapter 30](#)).

Proportional Assist Ventilation

Proportional assist ventilation (PAV) is another interactive form of noninvasive ventilation that adjusts the peak inspiratory pressure to the patient's work of breathing (10). This is accomplished by periodic measurements of respiratory mechanics (i.e., compliance and resistance), which are carried out (in a proprietary fashion) by ventilators capable of delivering PAV. There is no evidence that PAV is superior to PSV as a mode of noninvasive ventilation (11).

Face Masks

The emergence of noninvasive ventilation is largely due to advances in the design of the mask interface. Low-level CPAP can be delivered by nasal masks, but full face masks are required for BiPAP, and the face masks must be tight-fitting to minimize leaks (although most BiPAP machines can compensate for mask leaks). These face masks are uncomfortable (see [Figure 26.3](#)), and are poorly tolerated by patients. In addition to general discomfort, masks can cause skin breakdown at points of contact, especially on the bridge of the nose. "Mask intolerance" is a significant source of failed attempts to prevent intubation with noninvasive ventilation. In one study, 18% of the noninvasive ventilation failures were attributed to mask intolerance (12).



FIGURE 26.3 A tightly-secured face mask used for noninvasive ventilation.

The Helmet

An alternative to the tight-fitting face masks is available with a device called a “helmet”, which is a transparent hood that encloses the entire head, and has a soft collar that provides a seal at the neck. Studies comparing helmets and face masks have shown fewer intubations and even fewer deaths with the use of helmets (13). However, these devices are rarely used in adults.

USING NONINVASIVE VENTILATION

The utility and success of noninvasive ventilation (NIV) is determined by several factors, including the type of patient, the type and severity of illness, and the type of respiratory failure.

Type of Patient

The patient selection criteria for NIV are presented as a checklist in [Table 26.1](#). As a general rule, all patients with acute respiratory failure are candidates for NIV if the following criteria are satisfied: (a) the acute respiratory failure is not an immediate threat to life, (b) there is no life-threatening circulatory collapse, (c) the patient is awake or arousable, and is cooperative, (d) there are no persistent or recurrent seizures, (e) there is no hematemesis or persistent vomiting, and (f) there is no obstruction that will prevent effective ventilatory support through a face mask (e.g., laryngeal mass or edema). These criteria do not apply to patients who refuse intubation and mechanical ventilation, which has been reported in 20% of patients receiving NIV (14).

TABLE 26.1

Checklist for Noninvasive Ventilation

Are any of the following conditions present in a patient who presents with acute respiratory failure ?

	YES	NO
1. Agonal breathing.	☐	☒
2. Life-threatening circulatory collapse.	☐	☒
3. Severe agitation or uncontrolled seizures.	☐	☒
4. An acute confusional state.	☐	☒
5. Coma with inadequate airway protection.	☐	☒
6. Hematemesis or persistent vomiting	☐	☒

If the answer is NO to all of the above, the patient is a candidate for noninvasive ventilation.

NIV is a consideration in any case of acute respiratory failure that does not meet the criteria in Table 26.1. However, *the likelihood of success* (i.e., preventing intubation) is dependent on the *type of respiratory failure* (i.e., hypercapnic or purely hypoxemic respiratory failure), and the *underlying illness* (see next).

Hypercapnic Respiratory Failure

The most successful use of NIV is in patients with acute hypercapnic respiratory failure from an acute exacerbation of COPD (15). The benefits of NIV in this condition are shown in Figure 26.4. The graphs in this figure display the averaged results of 17 clinical studies, and they show that NIV drastically reduces the intubation rate, and has a smaller but significant survival benefit (15). As a result, *noninvasive ventilation is recommended as a first-line therapy for acute exacerbations of COPD that are associated with acute hypercapnic respiratory failure (4).* NIV does not have this benefit when the hypercapnia is chronic (4).

Evaluating the Response

About 10–20% of COPD patients with acute hypercapnia will not respond favorably to NIV (15,16). These patients can be identified by checking the PaCO_2 after one hour of NIV, and comparing it to the baseline PaCO_2 . *Failure of the PaCO_2 to decrease significantly (e.g., by at least 10%) after one hour of NIV is evidence of a poor response; i.e., NIV failure (16).* Altered mentation (e.g., from hypercapnia) can take longer to resolve (17), and should not be used as an early indication of NIV failure.

There is a reluctance to intubate patients with advanced COPD (because of the fear that mechanical ventilation will be a permanent condition), but delays to intubation can also have adverse consequences, and intubation should be a serious consideration at first evidence of NIV failure.

Obesity Hypoventilation Syndrome

Despite a paucity of studies, the available evidence shows that NIV is effective in reducing the intubation rate in patients with acute hypercapnic respiratory failure from the obesity hypoventilation syndrome (18). It seems reasonable to assume that the one-hour PaCO_2 evaluation just described is also valid for this condition.

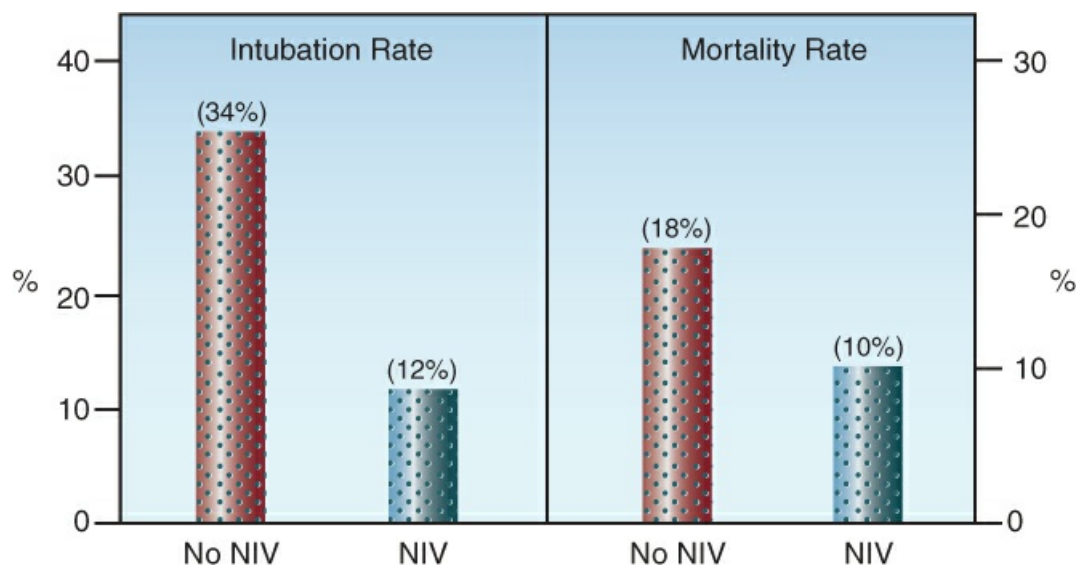


FIGURE 26.4 The effects of noninvasive ventilation on intubation rate and mortality rate in patients with acute exacerbation of COPD and hypercapnic respiratory failure. Data represents the pooled results from 17 clinical studies. From Reference 14.

Hypoxemic Respiratory Failure

The success of NIV in hypoxemic respiratory failure is dependent on the underlying condition. This is demonstrated in Figure 26.5, which shows the frequency of NIV failure in a variety of conditions that produce hypoxemic respiratory failure (19–21). The failure rate is lowest in cardiogenic pulmonary edema, and highest in community-acquired pneumonia. In the acute respiratory distress syndrome (ARDS), the likelihood of failure is dependent on the severity of illness (see also Figure 24.5) (20).

Evaluating the Response

Observations from clinical studies have shown that *failure to increase the $\text{PaO}_2/\text{FIO}_2$ ratio after one hour of NIV is evidence of a poor response; i.e., NIV failure* (19). This should be an indication for intubation if the hypoxemia is severe, or the underlying illness has a high failure rate for NIV.

Cardiogenic Pulmonary Edema

Both CPAP and NIV are highly successful (and equally effective) in cardiogenic pulmonary edema (22). This benefit is not limited to improvements in lung function, because *positive intrathoracic pressure can increase cardiac stroke output* (23,24). This is illustrated in Figure 26.6, which shows the changes in arterial pressure that occur during a positive pressure lung inflation. The positive airway (intrathoracic) pressure is accompanied by a steady rise in arterial pressure, which subsides as the pressure returns to baseline. The shaded areas in the arterial pressure waveforms (which are proportional to stroke volume) show that the increase in arterial pressure is a reflection of an increase in cardiac stroke output. This figure demonstrates that positive intrathoracic pressure increases the stroke output of the left ventricle. This effect is due to a decrease in left ventricular afterload (25), which is explained in the upcoming section on cardiac performance. (Note: The respiratory changes in stroke volume during mechanical ventilation can be used to evaluate fluid responsiveness, as shown in Figure 11.5).

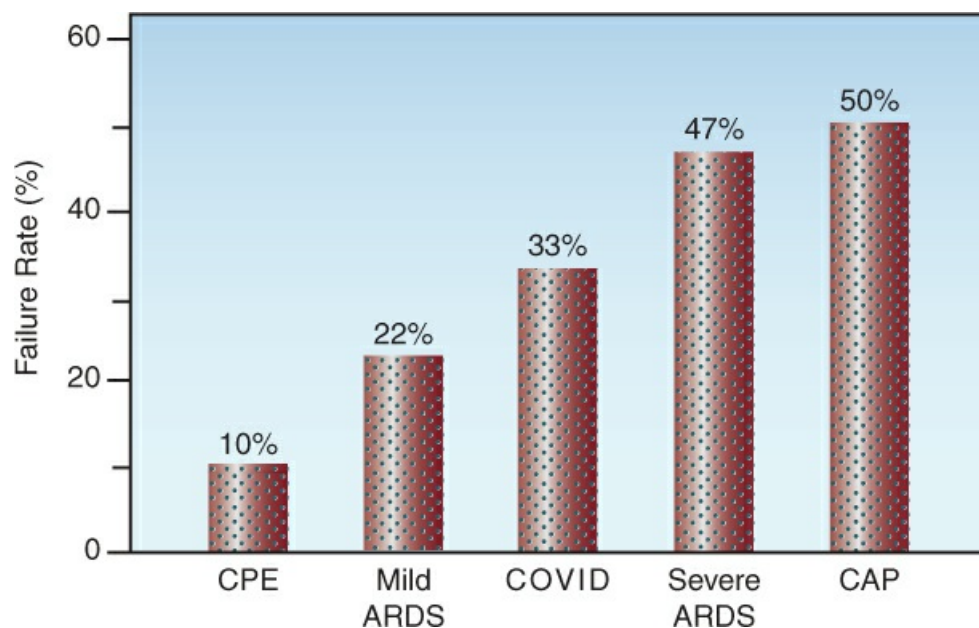


FIGURE 26.5 The failure rate of noninvasive ventilation in common conditions that cause acute hypoxemic respiratory failure. CPE = cardiogenic pulmonary edema, ARDS = acute respiratory distress syndrome, CAP = community acquired pneumonia. Data from References 19–21.

The Role of High-Flow Oxygen

High-flow nasal oxygen (HFNO), which is described in Chapter 25, is traditionally used for cases of hypoxemic respiratory failure when the hypoxemia is refractory to conventional methods of O₂ inhalation. However, HFNO has beneficial effects other than oxygenation, including an increase in lung compliance (distensibility), a decrease in the work of breathing, and a decrease in labored breathing (see Table 25.3) (26,27). These added benefits (which are attributed to the high flow rates) make HFNO a potential alternative to NIV for patients with hypoxemic respiratory failure, and this has been validated in studies showing equivalent intubation rates with HFNO and NIV in patients with acute hypoxemic respiratory failure (28). In addition, there is evidence that HFNO is superior to NIV for preventing intubation in immunocompromised patients (29).

The overall evidence indicates that *HFNO is equivalent to NIV for preventing intubation in*

patients with acute hypoxemic respiratory failure, and it may be more effective than NIV in immunocompromised patients. Considering that HFNO avoids the use of face masks, and thus is more comfortable for patients, while also allowing the oral intake of food and medications, it seems that HFNO should be preferred to NIV for patients with acute hypoxemic respiratory failure.

CARDIAC PERFORMANCE

The cardiorespiratory interactions in [Figure 26.6](#) demonstrate that the effects of positive pressure are not limited to the lungs. The cardiac effects of positive pressure breathing are complex, and involve the preload and afterload forces for the right and left sides of the heart ([30](#)).

Preload

The term “preload” refers to a force that stretches a resting muscle to a new length. An increase in preload (resting muscle length) increases the force of muscle contraction (as explained by the sliding filament model of muscle contraction). The preload force for the intact heart is the volume in the ventricles at the end of diastole, and *the Frank-Starling law of the heart* states that ventricular end-diastolic volume (preload) is the principal force that governs the strength of ventricular contraction ([31](#)). Thus, factors that affect preload can have a profound effect on cardiac output.

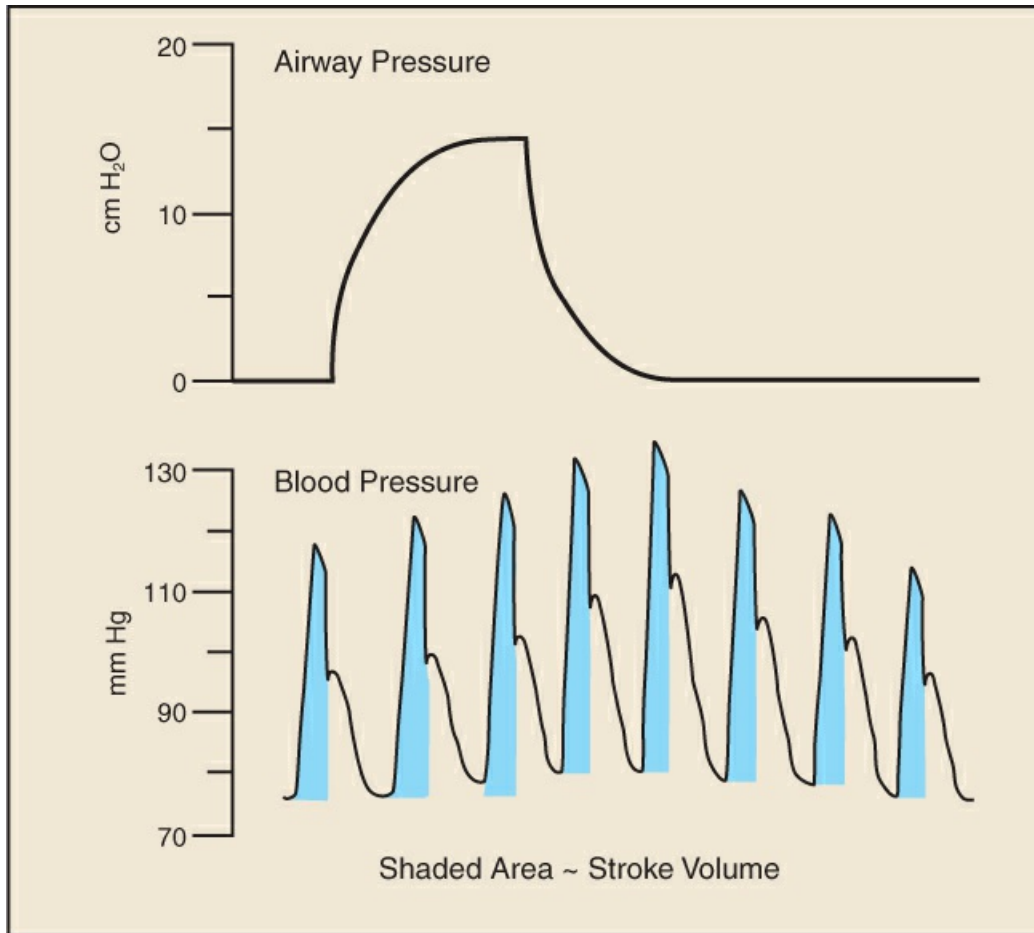


FIGURE 26.6 Changes in arterial pressure associated with a positive pressure breath. The increase in arterial pressure is a reflection of an increase in left ventricular stroke volume, as indicated by the shaded areas in the arterial pressure waveforms. See text for further explanation.

Positive Intrathoracic Pressure

Positive intrathoracic pressure can reduce ventricular preload (end-diastolic volume) in a number of ways. These are shown in [Figure 26.7](#), and are summarized below.

- . Positive intrathoracic pressure decreases the pressure gradient for venous inflow into the thorax, which decreases right ventricular filling.
- . The positive intrathoracic pressures used during mechanical ventilation (noninvasive and invasive) often exceed the pressures in the pulmonary veins, and this compresses the veins and impedes left ventricular filling.
- . Ventricular filling is a function of the end-diastolic *transmural* pressure (i.e., across the wall of the ventricles), so a positive pleural pressure will decrease this transmural pressure and thereby impede ventricular filling. This will have a greater effect on right ventricular filling, because the end-diastolic chamber pressure is lower in the right ventricle.
- . High intrathoracic pressures will increase pulmonary vascular resistance, and this can impede right ventricular outflow. In patients with right ventricular systolic dysfunction, this can

increase right ventricular volume and push the interventricular septum toward the left ventricle, thereby reducing left ventricular filling.

Positive pressure breathing is thus a real threat to cardiac filling. This threat is magnified by hypovolemia, so avoiding hypovolemia is imperative during positive pressure ventilation (especially when high pressures are required for lung inflation).

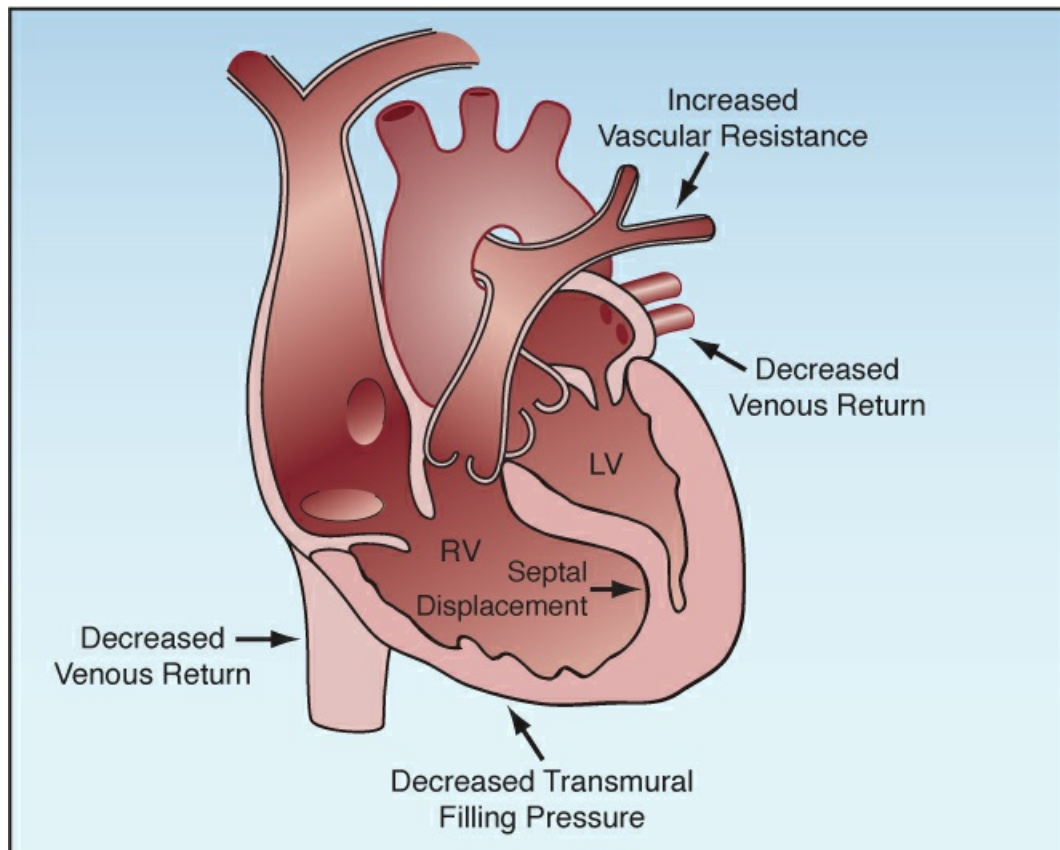


FIGURE 26.7 The different ways that positive intrathoracic pressure can decrease filling of the right and left ventricles.

Afterload

The term “afterload” refers to the force that must be overcome by muscle contraction. Unlike the preload force, which facilitates muscle contraction, the afterload force opposes muscle contraction. In the intact heart, *the afterload force is equivalent to the peak transmural wall pressure developed during systole*, which is the pressure needed to eject the stroke volume. This is a somewhat simplified view of afterload (which is more closely related to wall tension, and thus is dependent on wall thickness and chamber size), but it is adequate for describing the effects of positive pressure breathing.

Positive Intrathoracic Pressure

The determinants of left ventricular afterload are shown in [Figure 26.8](#). There is a tendency to view afterload as a sole function of impedance and resistance in the systemic circulation, but afterload is a *transmural pressure*, and thus is also a function of the pleural pressure surrounding

the heart (in the absence of a pericardial effusion): i.e.,

$$\text{Afterload} = \text{Chamber Pressure} - \text{Pleural Pressure} \quad (26.1)$$

The negative intrathoracic (pleural) pressure that occurs during spontaneous breathing will increase left ventricular afterload (by opposing the inward movement of the ventricular wall during systole), and this explains why the systolic blood pressure normally decreases during a spontaneous inspiration. An exaggeration of the inspiratory decrease in systolic blood pressure (>10 mm Hg) is a condition known as *pulsus paradoxus* (which is a misnomer, since the condition is not paradoxical), and this condition can be produced by exaggerated inspiratory efforts (32), as occurs in acute asthma.

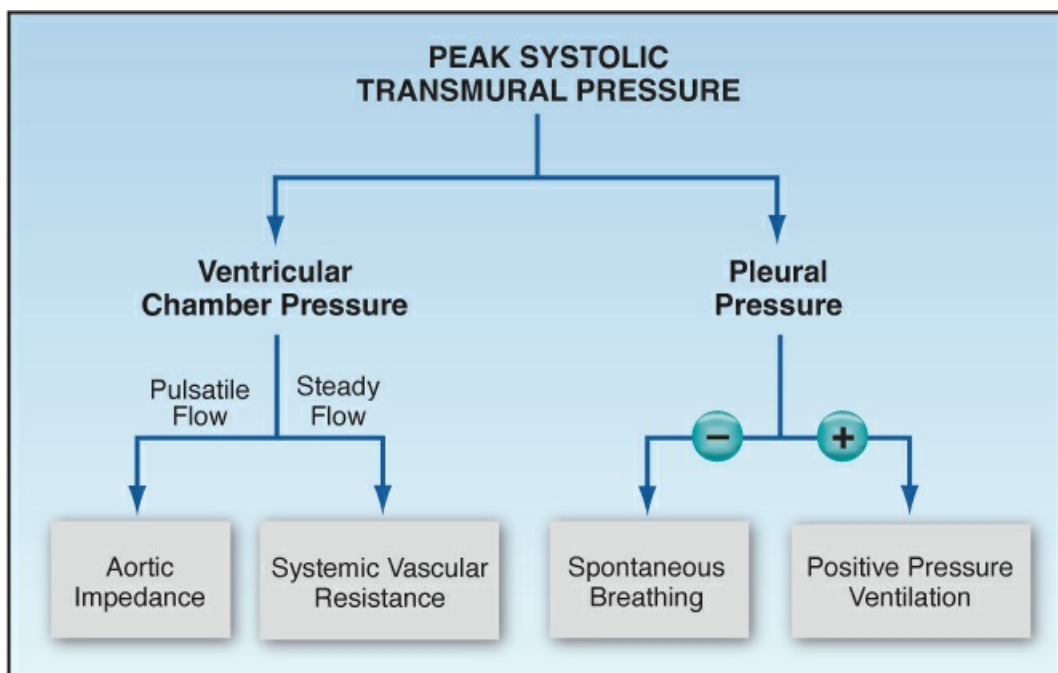


FIGURE 26.8 The determinants of left ventricular afterload. See text for explanation.

On the other hand, *positive intrathoracic pressure will decrease left ventricular afterload (by promoting the inward movement of the ventricle during systole)*, and this explains the inspiratory increase in systolic pressure in Figure 26.7. (This change is also known as *reversed pulsus paradoxus*.) Positive intrathoracic pressure thus assists the left ventricle in ejecting the stroke volume. This action is one of the proposed mechanisms for the ability of chest compressions to promote cardiac output (i.e., the thoracic pump model).

Cardiac Output

The overall effect of positive pressure ventilation on cardiac output will be determined by the balance between the preload and afterload effects of positive intrathoracic pressure, and this balance will be determined by three factors: the intravascular volume, the level of the intrathoracic pressure, and the presence or absence of cardiac dysfunction. The role of intravascular volume is mentioned previously, and the role of cardiac function is described next.

The Role of Cardiac Function

The following statements are based on the relationships between preload, afterload, and cardiac output in the normal and failing heart. These relationships are shown graphically in [Figure 16.2](#).

- . The normal heart operates on the steep portion of the preload curve and the flat portion of the afterload curve. In this situation, a decrease in preload has a greater influence on cardiac output than a decrease in afterload, so positive pressure ventilation is more likely to impair cardiac output by its actions to impede ventricular filling.
- . In contrast, the failing heart operates on the flat portion of the preload curve and the steep portion of the afterload curve. In this situation, a decrease in afterload has a greater influence on cardiac output than a decrease in preload, so positive pressure ventilation is more likely to promote cardiac output by its actions to reduce left ventricular afterload. This would explain the benefit of CPAP and NIV in cardiogenic pulmonary edema, as described earlier.

These statements have two important implications. First, in patients with normal cardiac function, it is essential to maintain intravascular volume to support venous return during positive pressure ventilation. Second, in patients with left ventricular dysfunction, positive intrathoracic pressure can promote cardiac stroke output as long as it does not impede venous inflow to the right heart.

A FINAL WORD

Don't Forget About Intubation

The popularity of noninvasive ventilation is based on the desire to avoid intubation, but there is a tendency to prolong noninvasive ventilation when there is no evidence of benefit, and this can be harmful. The following statements are relevant in this regard.

- . Evidence of benefit should be evident after one hour of noninvasive ventilation. This evidence includes a decrease in PaCO_2 in hypercapnic respiratory failure, an increase in the $\text{PaO}_2/\text{FIO}_2$ ratio in hypoxemic respiratory failure, and alleviation of respiratory distress.
- . Continued evidence of respiratory distress is particularly concerning, because it is often associated with increased spontaneous tidal volumes, and this can lead to *patient self-induced lung injury* (33) which is described in [Chapter 24](#) (see [Figure 24.4](#)), and has been implicated in the failure of noninvasive ventilation (34).
- . Delays to intubation creates unnecessary risk because emergent intubations in patients who are *in extremis* can be troublesome.

Although rarely viewed in a positive light, intubation and mechanical ventilation provide a greater level of ventilatory control, which may be necessary to slow or halt the progression of acute respiratory failure.

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A Final Word

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